MILITARY SPECIFICATION

COMPUTER, WEAPON AIMING
CP-1444/A

This Specification is approved for use by Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. The equipment, Computer, Weapon Aiming, CP-1444/A (WAC), covered by this specification is a microprocessor based digital controller. It accepts analog, digital, and discrete input signals from various systems within the aircraft and with ballistics data stored within the WAC, on programmable read only memory chips, performs the following:

a. Lead angle computations for the Head-Up-Display aiming of guns in Air-to-Air operational mode.
b. Continuously computed impact point computations for the head-up-display aiming of bombs, rockets, and guns in the Air-to-Surface operational modes.
c. Performs an interface function between the attitude and heading outputs of the attitude and heading reference system and the Signal Data Converter (SDC).
d. Performs an interface function between the Tacan Converter and the Signal Data Converter.
e. Performs an interface function between the UHF Homer Adapter and the Attitude Director Indicator (ADI).
f. Provides reference voltages and power supplies to other aircraft systems.

1.2 Associated equipment. This equipment shall operate with the associated equipment listed in 6.4.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: "Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (code 93) Lakehurst, New Jersey 08733", by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-W-5088 - Wiring; Aircraft, Selection and Installation of
MIL-E-5272 - Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-E-5400 - Electronic Equipment, Airborne, General Specification for
MIL-C-6021 - Castings, Classification and Inspection of
MIL-M-7793 - Meter, Time Totalizing
MIL-E-17555 - Electronic and Electrical Equipment Accessories, and Repair Parts
MIL-N-18307 - Nomenclature and Identification for Electronic, Aeronautical and Aeronautical Support Equipment Including Ground Support Equipment
MIL-T-23103 - Thermal Performance Evaluation, Airborne Electronic Equipment and Systems, General Requirement for
MIL-H-46855 - Human Engineering Requirements for Military Systems, Equipment and Facilities

STANDARDS

FEDERAL

FED-STD-595 - Colors

MILITARY

MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-704 - Aircraft Electric Power Characteristics
MIL-STD-781C - Reliability Design Qualification & Production Acceptance Tests Exponential Distribution
MIL-STD-1472 - Human Engineering Design Criteria for Military Systems, Equipment and Facilities
MIL-STD-2068 - Reliability Development Tests
MIL-STD-45662 - Calibration System Requirements

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.1.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

McDONNELL DOUGLAS CORPORATION

MDC A3780 - AV-8A/AV-8B/AV-8C/YAV-8B Vibration, Shock, and Acoustic Design Requirements and Test Procedures for Aircraft Equipment Installation.

(Application for copies should be addressed to the McDonnell Aircraft Company, Box 516, Saint Louis, Missouri 63166)

2.1.3 Order of precedence. When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

a. Contract. The contract shall have precedence over any specification.

b. This specification. This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications, where applicable, shall be specifically approved in writing by the procuring activity.

c. Referenced specifications. Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therin. All referenced specifications shall apply to the extent specified.

3. REQUIREMENTS

3.1 First article. When specified (see 6.3), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.3)

3.2 Parts and materials. In the selection of parts and materials fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

a. Microelectronic integrated circuits shall be used as a prime functional element of the equipment.

b. Circuits shall be designed to use a minimum of different parts, subassemblies, cards and modules.

c. Parts and materials shall be procured from Qualified Product List (QPL) sources when they exist.

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d. Nonrepairable subassemblies, as outlined in MIL-E-5400, shall only be used when practicable.

e. When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable subassemblies without the approval of the procuring activity.

3.2.1 Nonstandard parts. Nonstandard parts and materials (including electron tubes, transistors and diodes) other than microeletronic devices shall be obtained using MIL-E-5400 as a guideline.

3.2.2 Modules. The equipment shall be functionally modularized at all levels of assembly/disassembly. The Weapons Replaceable Assembly (WRA) shall be packaged in modular form easily removable for repair by replacement.

3.2.3 Finish and colors. The equipment shall be finished in accordance with MIL-E-5400. The case shall be lusterless Black, Color No. 37038, in accordance with FED-STD-595, to provide enhancement of heat transfer.

3.2.4 Castings. Alloy castings used in the design and construction of the equipment shall be in accordance with MIL-E-5400. The classification, grade, and inspection requirements of MIL-C-6021 are applicable.

3.2.5 Dissimilar metals. Grouping of similar and dissimilar metals shall be in accordance with MIL-E-5400.

3.3 Design and construction. The equipment shall conform with all the applicable requirements of MIL-E-5400 Class 2 for design and construction as specified herein.

3.3.1 Total weight. The total weight of the equipment, excluding external cables, shall not be greater than 18 pounds.

3.3.2 Reliability

3.3.2.1 Operational stability. The equipment shall meet the performance test requirements when operated continuously or intermittently, for a period of at least 1,000 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating life. The equipment shall have a minimum total operating life of 10,000 hours with depot servicing and replacement of parts.

3.3.2.3 Storage life. The equipment shall meet the performance and service life requirements of this specification, without component or part replacement, readjustment, or other maintenance action (excluding desiccant replacement) after 18 month storage period under any construction of climatic environments specified herein.

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3.3.2.4 Reliability in mean time between failure (MTBF). The equipment, including any Built-in-Test provision, shall have 1,000 hours of mean (operating) time between failures with a minimum acceptable level of 600 hours mean (operating) time between failures.

3.3.2.5 Time totalizing meter. The equipment shall contain an elapsed time totalizing meter which meets the requirements of meter type MS2765 in accordance with MIL-M-7793. The meter shall be located such that it may be read without the removal of a cover or any other part from the unit. The meter shall indicate "ON" time which is any operational mode.

3.3.3 Cabling and connections

3.3.3.1 Cabling and connectors. The equipment shall provide for the use of cables and connectors in accordance with MIL-E-5400 except as specified herein.

3.3.3.2 Interconnection cabling. The equipment shall be capable of meeting the performance test requirements using external wiring in accordance with MIL-W-5088. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

3.3.3.3 Connectors. Connectors for joining of external aircraft wiring to the equipment shall be in accordance with 3.5.1.10.

3.3.4 Control panel. Not applicable.

3.3.5 Interchangeability. In addition to the requirements of MIL-E-5400, all contractor parts, subassemblies, and WRA's having the same part number shall be interchangeable with each other with respect to installation and performance.

3.3.6 Interference control. The equipment shall be in accordance with MIL-STD-461 as specified herein.

3.3.6.1 Conducted emission. The conducted emissions of the equipment shall be controlled within the limits and requirements of 4.3.3.3.1.

3.3.6.2 Conducted and radiated susceptibility There shall be no malfunction, change in normal indication or degradation of performance of the equipment when subjected to the tests and requirements of 4.3.3.3.1.

3.3.7 Maintainability. The design, configuration, installation, and operation of the equipment shall minimize servicing requirements, including the need for special tools and support equipment, personnel skills, and manpower. The design shall permit operational verification of the equipment at the organizational level and maintenance/performance testing at the depot level.

3.3.7.1 Organization Level Maintenance-To-Repair. Organizational level maintenance, using perodic built-in-test programs and initiated pilot confidence check program functions, and excluding access and close up (function of aircraft installation), shall be not more than 0.5 direct man-hours mean-time-to-repair (MTTR).

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3.3.7.2 Human performance/human engineering. The equipment shall be designed in accordance with MIL-STD-1472, MIL-H-46855 and the criteria specified herein.

3.3.7.2.1 Functional packaging. The equipment shall be packaged on a functional basis to minimize the time required to conduct performance tests and isolate failures. SRA guides, connector keying, and clear, unambiguous marking shall be provided to simplify replacement.

3.3.7.2.2 Extractors. Shop replaceable Assemblies (SRA's) shall contain extractors which utilize mechanical leverage to simplify SRA withdrawal and replacement. It shall not be necessary to restrain the chassis in order to remove SRA's.

3.3.7.2.3 Handling equipment. The computer shall be provided with an integral external handle to permit transport to and from the aircraft with no requirement for a handling fixture.

3.3.7.2.4 Control and adjustment access. Controls that should not be adjusted after installation shall be inaccessible when installed in the next higher assembly.

3.3.7.3 Maintenance concepts. Design shall be such that maintenance can be performed under the following concepts:

3.3.7.3.1 Preventive maintenance. The equipment shall not have preventive maintenance requirements.

3.3.7.3.2 Scheduled maintenance. Scheduled maintenance shall consist of preflight and post flight maintenance, performed at the organizational maintenance level.

3.3.7.3.3 Unscheduled maintenance.

3.3.7.3.3.1 Organizational maintenance level. Consists of verifying and isolating the fault to the Weapon Aiming Computer, removing and installing a like serviceable unit and performing operational checks to verify proper unit operation.

3.3.7.3.3.2 Intermediate maintenance level. The repair and maintenance of unserviceable WRA's shall be the function of depot maintenance level.

3.3.7.3.3.3 Depot maintenance level. Consists of bench checks, disassembly and repair of unserviceable WRA's by replacement of discrepant subassemblies. Serviceability of all repaired items will be verified (performance test) prior to return to serviceable stocks.

3.3.7.3.3.4 Semi-Automatic checkout equipment. The unit shall be compatible with semi-automatic checkout equipment when used in conjunction with suitable interfacing equipment and shall include test provisions to permit performance verification testing at depot level maintenance.

3.3.7.3.4 Test points. Test points shall be exposed and grouped within a multipin connector which is readily accessible at the level of maintenance for which it is intended.

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3.3.7.3.4.1 External test points. Each unit shall have sufficient accessible external test points that when used in conjunction with the operating connector, will permit quantitive determination of the performance level of the WRA and if substandard performance is determined, permit isolation of the fault to an SRA function.

3.3.7.3.4.2 Internal test points. Each level of disassembly within the WRA shall expose accessible internal test points when disassembled to the extent necessary for depot level maintenance.

3.3.7.3.5 Self test. The unit shall contain both periodic and initiated built-in self-test capabilities to permit automatic and semiautomatic fault detection and isolation to the WRA.

3.3.7.3.5.1 Periodic bit. Periodic bit shall automatically operate beginning with the application of power (subject to warm-up time) to the equipment. Periodic bit shall not affect the WRA/HUD SYSTEM. The Bit feature shall allow for maintenance at the organizational level and assist in isolating faults to the WRA while installed in the aircraft.

3.3.7.3.5.2 Initiated bit. The unit shall have the capacity provided to place the computer in a self-test mode (Pilot's Confidence Check) by operation of an external control. In this mode the computer will not be required to perform other functions. In the self-test mode, the computer via the Signal Data Converter, shall be capable of displaying a test pattern on the Digital Display Indicator (DDI) that will indicate to the operator that the Weapon Aiming Computer is operating in accordance with 3.4.3.

3.3.7.3.5.2.1 Air-to-surface initiated bit. The equipment shall display on the DDI, a predetermined measureable continuously computed impact point position, when in the Weapons 'A' or Weapons 'B' operational mode, with the Pilot's Confidence Check switch depressed.

3.3.7.3.5.2.2 Air-to-air initiated bit. The equipment shall display on the DDI, a predetermined measurable aiming dot (with variable stadiametric ranging symbol) when in the air-to-air operational mode, with the Pilots' Confidence Check switch depressed.

3.3.7.3.5.3 Fault indicator. A mechanical fault indicator shall be provided on the WRA. The fault indicator shall be energized in the event a malfunction is detected by the periodic test circuitry. Once set under program control, the indicator shall be visible after removal of power.

3.3.7.3.5.4 Pilot warning indicator. The equipment shall provide a 28 Volt DC potential to the centralized warning panel in the event a malfunction is detected by the WAC periodic test circuitry.

3.3.7.3.5.5 Weapon aiming computer bit indicator. The equipment shall have a LED indicator which shall provide current BIT testing status. When illuminated, equipment readiness is indicated.

3.3.8 Identification of equipment. Nameplate, identification markings, and nomenclature for the unit shall be in accordance with MIL-N-18307.

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3.3.9 Standard conditions. The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

a.	Temperature	Room ambient (25°C ± 5°C)
b.	Vibration	None
c.	Humidity	Room ambient up to 90 percent relative humidity
d.	Input Power Voltage	115/200 VAC +3 percent -6 percent, 3 phase. 400 Hz ± 4Hz 28 V ± 1.0 VDC

3.3.10 Service conditions. The unit shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions in accordance with MIL-E-5400 for the class of equipment defined herein and except as modified in paragraphs 3.3.10.1 through 3.3.10.4.

Weapon Aiming Computer	2	III	Continuous	-54°C to +71°C
			Nonoperation	+71°C to +95°C

3.3.10.1 Vibration. The equipment shall operate when subjected to random vibration as described by "Shelf Only", Table III, entitled, "Random Vibration Test Levels, Region C", of McDonnell Aircraft Company Design Requirements Publication, MDC A3780. The vibration tolerance shall be as follows:

10 - 500 Hz +3, -1.5 dB
500 - 1000 Hz ±3 dB Deviations of ±6 dB shall be allowable over cummulative bandwidth of 100 Hz.

3.3.10.2 Temperature/altitude. The equipment shall operate satisfactorily when subjected to the temperature/altitude requirements of MIL-E-5400 Table 1, Class 2. Temperature/Altitude conditions applicable for continuous and intermittent operation shall be as defined in Figure 1. The equipment shall be capable of satisfactory operation following exposure to a -54°C environment for 2 hours and a +95°C environment for 16 hours.

3.3.10.3 Acceleration. The equipment shall operate when subjected to the acceleration inputs in accordance with MIL-E-5272 Procedure III.

3.3.10.4 Non-operating. The equipment shall be capable of withstanding prolonged exposure to temperature extremes in accordance with MIL-E-5400, Table 1.

3.3.11 Warm-up time. The time required for the unit to warm up prior to operation shall be kept to a minimum and shall be not greater than two minutes under the aircraft installed environment as specified herein.

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3.3.12 Reserve processing time. Total system frametime shall contain reserve processing time of at least 20 percent of the total system frametime.

3.3.13 Input power requirements. The unit shall meet all applicable requirements of MIL-STD-704 for Category B equipment and shall provide specified performance from power sources with characteristics as defined in MIL-STD-704. The power required shall be not greater than the following amounts:

a.	AC Power	115/200 VAC 3 phase 400 Hz Category B 120 VA at .9 PF
b.	DC Power	28 VDC Category B Up to 1.0 Amps.

3.3.13.1 Under voltage protection. The computer shall not be damaged by voltages below the minimum specified herein and shall automatically resume normal operation when the voltage returns within limits.

3.3.13.2 Power drop out. The equipment shall automatically recover after a momentary power interruption upon resumption of normal power within the following periods.

	Power Interruption Time	Maximum Recovery Time
a.	0 to .005 seconds	Immediate
b.	.005 to 1.0 seconds	3.0 seconds
c.	1.0 to 5.0 seconds	5.0 seconds
d.	5.0 to 10.0 seconds	15.0 seconds
e.	10.0 seconds or greater	30.0 seconds

3.3.14 Cooling.

3.3.14.1 Thermodynamic design. The thermal design of the equipment shall be in accordance with MIL-E-5400 and MIL-T-23103 except as otherwise specified herein. Each thermal interface within the equipment shall be designed to maintain the necessary thermal conduction over the life of the equipment. The design shall consider the material type, flatness, hardness, spring rate, fastener torquing, and fatigue strength. In addition, the conductive heat transfer area, interface force, and dimensional variation of the thermal interface between plug-in SRA's and the next higher level of assembly shall be considered in the design.

3.3.14.2 Heat dissipation. The heat generated by the equipment shall be dissipated by convection to the installed environment, conduction to mounting structure and radiation to the environment.

3.4 Performance. Values set forth to establish the requirements of satisfactory performance apply to performance under both standard and extreme service and input power conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions shall be specified in the Contract, unless otherwise specified.

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3.4.1 Operation. The equipment shall function as an integral part of a weapon aiming system which accepts digital and analog sensor information and processes them to provide, appropriate lead angle computation for air-to-air gun aiming, air-to-air target range, determination of impact point of a released air-to-surface weapon in either, Weapon A mode (no wind solution) or Weapon B mode (cross heading wind correction), advanced breakaway warning, breakaway warning, air-to-ground target height, pilot's confidence check in either air-to-air or air-to-ground mode.

3.4.2 Modes of operation. The Weapon Aiming Computer shall provide the followng operational modes.

3.4.2.1 Air-to-surface mode. Utilizing externally provided signals (digital and analog) from aircraft sensors, the Weapon Aiming Computer shall compute Azimuth Steering Line (ASL) angle and length, and Continuously Computed Impact Point (CCIP) azimuth and elevation signals for the Head-Up Display aiming and manual release of rockets, guns, low-drag and high-drag bombs in the air-to-surface attack modes.

3.4.2.1.1 Breakaway warning. A flashing breakaway warning signal shall be generated for display on the Head-Up Display when the aircraft is not above minimum release altitude defined by the hazards of collision with surface or of the aircraft penetration of weapon fragmentation zone or both.

3.4.2.1.2 Advanced breakaway warning cue. In Weapons A or B mode the computer shall generate an advanced warning cue signal that will travel along the azimuth steering line (ASL) at a rate such that it will terminate at the continuous computed impact point cue when breakaway warning occurs.

3.4.2.1.3 Occurrence of breakaway warning and advanced breakaway warning signals. Breakaway warning and advanced breakaway warning shall be displayed only when the gun safety switch or bomb safety switch is activated.

3.4.2.2 Air-to-air mode. Utilizing externally provided signals (digital and analog) from aircraft sensors, the Weapon Aiming Computer shall compute azimuth and elevation lead angle signals for the HUD aiming of the 30 mm guns in the air-to-air attack mode.

3.4.3 Pilots confidence check (PCC).

3.4.3.1 Air-to-surface PCC. The Weapon Aiming Computer shall in Weapons A or B operational mode, and with the WAC test switch manually activated, apply (fixed) voltages to the input circuitry of the WAC unit for computation of a predetermined Continuously Computed Impact Point and Azimuth Steering Line which shall be displayed on the Head-Up Display for evaluation by the operator for unit acceptability.

3.4.3.2 Air-to-air PCC. The Weapon Aiming Computer shall in Air-to-Air operation mode, and with the WAC test switch manually activated, apply (fixed) voltages to the input circuitry of the WAC unit for computation of a predetermined Air-to-Air aiming mark which shall be displayed on the Head-Up Display for evaluation by the operator for unit acceptability.

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3.4.4 Reset implement. The computer shall receive a reset implement signal from the Signal Data Computer unit. The signal shall be a positive 2.0 millisec ± 2 micro sec pulse, TTL compatible, recurring at a rate of 40.69 Hz. The reset pulse shall be utilized to synchronize the start of the WAC with the SDC unit.

3.4.5 Built-in-test (BIT). BIT shall be contained within the WAC to accomplish failure detection for at least 90 percent of all equipment faults with a 98 percent confidence level.

3.4.5.1 BIT operating modes. BIT shall operate in two modes, periodic and initiated. These modes shall provide inflight, shipboard, and ground failure detection and isolation of WAC failures. Periodic BIT shall automatically operated beginning with the application of power (subject to the warm-up time) to the equipment. Initiated BIT shall function in response to a discrete signal from the WAC Test Switch (Pilots Confidence Check) located on the center console.

3.4.5.2 BIT performance. Periodic BIT shall detect 76 percent of the WAC failures. Initiated BIT shall detect 14 percent of WAC failures. The sum total of Periodic and Initiated BIT shall total 90 percent. Any malfunction (including a false alarm) of the Built-In-Test features of the equipment which exceeds the equipment requirements for Built-In-Test shall be classified as a relevant failure.

3.4.5.3 BIT test times. Periodic BIT shall perform the specified functions once every display refresh time. Initiated BIT shall provide a continuous test pattern when Pilots Confidence Test is selected.

3.4.5.4 BIT failure indicator. A BIT failure indicator shall provide a distinct indication of the status of the WAC. The indicator shall be tripped to the latched (fail) position in response to a failure signal generated by the WAC and shall remain latched until reset by maintenance action. An LED indicator shall also be included to provide a "GO" indication in the event an intermittent condition actuates the the BIT indicator.

3.4.6 Universal asynchronous receiver transmitter (UART). Two UART's operating in tandem shall provide for 16 bits of parallel data to be converted to two serial 8 bit data streams and sequentially transmitted by a differential line driver to the Signal Data Converter. The serial format of the data shall be 1 start bit, 1 stop bit, 8 data bits, and odd parity transmitted at a serial data rate of 125K Hz. The associated data that is passed between the WAC and the SDC shall be as listed in Table I.

3.4.7 Ballistics code. The computer shall store ballistics data for twenty-one (21) different types of weapons. Ballistics appropriate to the weapon being released shall be selected on the Ballistic Selector Panel and shall be connected to the WAC unit by a six wire binary encoded signal line (See Table II).

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TABLE I. WAC/SDC transmitted data.

WAC Transmissions
Bank Angle (COS)
Bank Angle (SIN)
Heading (Angle)
Heading Sector
Elevation (Pitch Angle)
Tacan (Angle)
True Air Speed
Aiming Mark Elevation
Aiming Mark Azimuth
Ballistics Not Valid
CCIP Cue Elevation
CCIP Cue Azimuth
Az Steering Line (Angle)
Az Steering Line (Length)
Reflection Length
Breakaway
Stadiametric Range
Target Height
Advance Breakaway Warning Length

SDC Transmissions	
Data 1-Baro Ht A (Uncorrected, Unscaled)	
Data 2-Baro Ht B (Corrected)	
Data 3-BIT 00 to BIT 12	Radio HT (0 to 5000ft)
BIT 13	1=Wpn A 0=Not WPN A
BIT 14	0=Radio Valid 1=Radio Not Valid
BIT 15	1= Baro Select 0=Radio Select

3.4.8 Power supply. All DC voltages required to operate the computer shall be contained in the unit. The voltages are as follows: +5; +10, -10; +15, -15; +20, -20; volts DC.

3.4.9 Reference voltages. The unit shall provide isolated supply voltages to the following aircraft sensors.

a. Ballistics Selector Panel
b. Weapon Control Panel
c. Vertical Speed Transducer
d. Air Data Computer
e. Target Height Set Potentiometer

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TABLE II. Ballistics code.

Weapon	BSP Code
1. MK.76 PRACTICE BOMB	010100
2. MK.81 LDGP BOMB	110101
3. MK.81 SNAKEYE LD BOMB	110000
4. MK.82 LDGP BOMB	010111
5. MK.82 SNAKEYE LD BOMB	010001
6. MK.83 LDGP BOMB	010010
7. MK.106 PRACTICE BOMB	011011
8. MK.77 FIRE BOMB	101000
9. MK.81 SNAKEYE HD BOMB	011000
10. MK.82 SNAKEYE HD BOMB	111001
11. CBU MK.20 ROCKEYE 1.2 SEC FUZE CLUSTER BOMB	000011
12. CBU MK.20 ROCKEYE 4.0 SEC FUZE CLUSTER BOMB	100001
13. CBU MK.59/B APAM 1.2 SEC FUZE CLUSTER BOMB	011101
14. CBU MK.59/B APAM 4.00 SEC FUZE CLUSTER BOMB	110110
15. 2.75 IN FFAR MK.1 OR MK.5 ROCKET	111100
16. 2.75 IN FFAR MK.151 ROCKET	100100
17. 5.0 IN FFAR ZUNI ROCKET GROUP 1	000110
18. 5.0 IN FFAR ZUNI ROCKET GROUP 2	000101
19. 5.0 IN FFAR ZUNI ROCKET GROUP 3	100111
20. 5.0 IN FFAR ZUNI ROCKET GROUP 4	100010
21. 30 MM ADEN GUN	000000

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3.5 Detail Requirements.

3.5.1 Weapon Aiming Computer. The Weapon Aiming Computer henceforth referred to as the computer shall meet the following requirements.

3.5.1.1 Function. The computer, when connected to the associated equipment listed in 6.4 shall receive AC, DC analog, discrete, 3 wire synchro, and digital data signals; process this data to perform the functions specified in 3.4.1.

3.5.1.2 Organization. The Central Processor shall be a stored programmed fixed point, custom-designed, microcontroller computer. Integer data representation shall be two's complement. The operand word length shall be 16 bits for single precision arithmetic and 32 bits for floating point arithmetic. Floating point representation in the Arithmetic Processor is 32 bits, 1 for sign, 7 for exponent, and 24 for mantissa.

3.5.1.3 Program storage memory. This type of memory shall be Programmable Read Only Memory (PROM), organized as 2048 locations x 8 bits wide. Total addressable PROM memory shall be capable of 16,384 words of 16 bits each.

3.5.1.4 Read/write memory. This type of memory shall be random access memory (RAM) organized as 256 locations x 4 bits wide. Total RAM memory shall be capable of 1280 words of 16 bits each.

3.5.1.5 Data protection. The computer shall contain the necessary circuitry to protect memory data from any normal or abnormal input power transients. When normal power is restored, a restart sequence shall cause the computer to return to a start-up point in the program. Data from external sources that would have arrived during the power interruption will be lost.

3.5.1.6 Floating point operation. Floating point operation shall be performed by either of two Arithmetic Processing Units (APU) which provide high performance 32 bit floating point arithmetic and floating point trigonometric and mathematical operations including square roots.

3.5.1.7 Form factor. The computer shall conform to the dimensions specified in Figure 2.

3.5.1.8 Weight. The computer shall be not greater than 18 pounds.

3.5.1.9 Contents. The computer shall contain the following functional sections.

a. Input Interface
b. Central Processor
c. Arithmetic Processor
d. Ouput Interface
e. Power Supply

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3.5.1.10 Connections. The connections to/from the computer shall be in accordance with Table III for Aircraft Interconnect, Table IV for Test Connector Interconnect and are designated as follows:

	Designation	Function	Receptacle Type
a.	J1A/B	Aircraft Interconnect	M81659/35A2-0035
b.	J2	Test Connector	M24308/4-16

TABLE III. Aircraft interconnect.

CONNECTOR PIN	SIGNAL NAME	SOURCE*	DESTINATION
J1A - 1	EARTH BONDING		AIRCRAFT EARTH BONDING
J1A - 2	SYSTEM ON (28V)		VERTICAL RATE GYRO
J1A - 3	EARTH SCREEN	WAC	
J1A - 4	HUD ON (28V)	DSC	
J1A - 5	OV RETURN (28V)	AIRCRAFT POWER	
J1A - 6	-		
J1A - 7	SYSTEM ON (28V)		HORIZ. RATE GYRO
J1A - 8	RESET IMPLEMENT	SDC	
J1A - 9	AIR-TO-AIR ON (28V)	DSC	
J1A - 10	STICK INTERVAL (+ts)	WCP	
J1A - 11	STICK LENGTH (-nts)	WCP	
J1A - 12	PARITY BALLISTICS	BSP	
J1A - 13	2^0 BALLISTICS	BSP	
J1A - 14	2^1 BALLISTICS	BSP	
J1A - 15	2^2 BALLISTICS	BSP	
J1A - 16	2^3 BALLISTICS	BSP	
J1A - 17	2^4 BALLISTICS	BSP	
J1A - 18	-		
J1A - 19	-		
J1A - 20	-		

*See paragraph 6.4

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TABLE III. Aircraft interconnect. - Continued

CONNECTOR PIN	SIGNAL NAME	SOURCE*	DESTINATION
J1A -21	WEAPONS 'B' ON (IN)	DSC	
J1A -22	WEAPONS 'B' ON (OUT)		SDC
J1A -23	-		
J1A -24	-		
J1A -25	-		
J1A -26	-		
J1A -27	-		
J1A -28	UHF HOMER (+ve)		ADI
J1A -29	UHF HOMER (-ve)		ADI
J1A -30	TARGET HEIGHT	CCP	
J1A -31	1/RANGE	TWISTGRIP VIA DSC	
J1A -32	UHF HOMER (+ve)	HOMER ADAPTER	
J1A -33	UHF HOMER (-ve)	HOMER ADAPTER	
J1A -34	OV DC REF		ADC
J1A -35	+10V DC REF		ADC
J1A -36	+5V DC REF (LOGIC)		BSP
J1A -37	OV DC REF (LOGIC)		BSP
J1A -38	+10V DC REF		WCP
J1A -39	OV DC REF		WCP
J1A -40	-10V DC REF		WCP

*See paragraph 6.4

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TABLE III. Aircraft interconnect. - Continued

CONNECTOR PIN	SIGNAL NAME	SOURCE*	DESTINATION
J1A -41	-		
J1A -42	-		
J1A -43	-		
J1A -44	-		
J1A -45	-		
J1A -46	-		
J1A -47	-		
J1A -48	-		
J1A -49	-		
J1A -50	-		
J1A -51	-		
J1A -52	+15V DC REF		VST
J1A -53	OV DC REF		VST
J1A -54	-15V DC REF		VST
J1A -55	-		
J1A -56	-		
J1A -57	-		
J1A -58	-		
J1A -59	-		
J1A -60	-		

*See paragraph 6.4

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TABLE III. Aircraft interconnect. - Continued

CONNECTOR PIN	SIGNAL NAME	SOURCE*	DESTINATION
J1A -61	-		
J1A -62	INDICATED AIRSPEED	ADC	
J1A -63	TRUE AIRSPEED	ADC	
J1A -64	BARO HEIGHT	ADC	
J1A -65	BARO DATUM (SET)	SDC	
J1A -66	ANGLE OF ATTACK	ASDD	
J1A -67	-		

*See paragraph 6.4

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TABLE III. Aircraft interconnect. - Continued

CONNECTOR PIN	SIGNAL NAME	SOURCE*	DESTINATION
J1B - 1	+10V DC REF		CCP
J1B - 2	OV DC REF		CCP
J1B - 3	EARTH SCREEN	WAC	
J1B - 4	-		
J1B - 5	-		
J1B - 6	TX + DATA BYTE 2		SDC
J1B - 7	TX - DATA		
J1B - 8	PILOTS CONFIDENCE CHECK	WAC TEST SWITCH (CTR CONSOLE)	
J1B - 9	REC. + DATA BYTE 1	SDC	
J1B - 10	REC. - DATA		
J1B - 11	TX + DATA BYTE 1		SDC
J1B - 12	TX - DATA		
J1B - 13	STORE HEADING (28V)	CONTROL COLUMN	
J1B - 14	WAC FAIL (28V)		CWP
J1B - 15	-		
J1B - 16	-		
J1B - 17	-		
J1B - 18	TACAN SYNCHRO X (S1)	TACAN CONVERTER	
J1B - 19	Y (S3)		
J1B - 20	Z (S2)		

*See paragraph 6.4

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TABLE III. Aircraft interconnect. - Continued

CONNECTOR PIN	SIGNAL NAME	SOURCE*	DESTINATION
J1B -21	PITCH SYNCHRO Y (S3)	ECA	
J1B -22	X (S1)		
J1B -23	Z (S2)		
J1B -24	ROLL SYNCHRO X (S1)	ECA	
J1B -25	Y (S3)		
J1B -26	Z (S2)		
J1B -27	HEADING SYNCHRO Y (S3)	ECA	
J1B -28	X (S1)		
J1B -29	Z (S2)		
J1B -30	REC. + DATA BYTE 2	SDC	
J1B -31	REC. - DATA		
J1B -32	A PHASE 400 Hz	AIRCRAFT POWER	
J1B -33	B PHASE 400 Hz	AIRCRAFT POWER	
J1B -34	C PHASE 400 Hz	AIRCRAFT POWER	
J1B -35	NEUTRAL	AIRCRAFT POWER	
J1B -36	-		
J1B -37	ROLL RATE +	HORIZ. RATE	
J1B -38	ROLL RATE -	GYRO	
J1B -39	YAW RATE +	VERTICAL RATE	
J1B -40	YAW RATE -	GYRO	

*See paragraph 6.4

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TABLE III. Aircraft interconnect. - Continued

CONNECTOR PIN	SIGNAL NAME	SOURCE*	DESTINATION
J1B -41	PITCH RATE +	VERTICAL RATE	
J1B -42	PITCH RATE -	GYRO	
J1B -43	-		
J1B -44	-		
J1B -45	-		
J1B -46	-		
J1B -47	-		
J1B -48	-		
J1B -49	-		
J1B -50	MACH NO.	SDC	
J1B -51	WEIGHT ON WHEELS	MAIN LANDING GEAR	
J1B -52	-		
J1B -53	-		
J1B -54	-		
J1B -55	-		
J1B -56	-		
J1B -57	-		
J1B -58	-		
J1B -59	-		
J1B -60	-		

*See paragraph 6.4

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TABLE III. Aircraft interconnect. - Continued

CONNECTOR PIN	SIGNAL NAME	SOURCE*	DESTINATION
J1B -61	-		
J1B -62	-		
J1B -63	-		
J1B -64	-		
J1B -65	-		
J1B -66	-		
J1B -67	-		

*See paragraph 6.4

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TABLE IV. Test connector.

CONNECTOR PIN	SIGNAL NAME	
J2 - 1	DB00	DATA BUS
J2 - 2	DB01	DATA BUS
J2 - 3	DB02	DATA BUS
J2 - 4	DB03	DATA BUS
J2 - 5	DB04	DATA BUS
J2 - 6	DB05	DATA BUS
J2 - 7	DB06	DATA BUS
J2 - 8	DB07	DATA BUS
J2 - 9	DB08	DATA BUS
J2 - 10	DB09	DATA BUS
J2 - 11	DB10	DATA BUS
J2 - 12	DB11	DATA BUS
J2 - 13	DB12	DATA BUS
J2 - 14	DB13	DATA BUS
J2 - 15	DB14	DATA BUS
J2 - 16	DB15	DATA BUS
J2 - 17	9MHz	OSCILLATOR OUT
J2 - 18	3MHz	CLOCK OUT
J2 - 19	2MHz	CLOCK OUT
J2 - 20		

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TABLE IV. Test connector. - Continued

CONNECTOR PIN	SIGNAL NAME	
J2 - 21		
J2 - 22	AB01	ADDRESS BUS
J2 - 23	AB02	ADDRESS BUS
J2 - 24	AB03	ADDRESS BUS
J2 - 25	AB04	ADDRESS BUS
J2 - 26	AB05	ADDRESS BUS
J2 - 27	AB06	ADDRESS BUS
J2 - 28	AB07	ADDRESS BUS
J2 - 29	AB08	ADDRESS BUS
J2 - 30	AB09	ADDRESS BUS
J2 - 32	AB10	ADDRESS BUS
J2 - 33	AB11	ADDRESS BUS
J2 - 34	AB12	ADDRESS BUS
J2 - 35	AB13	ADDRESS BUS
J2 - 36	AB14	ADDRESS BUS
J2 - 37	AB15	ADDRESS BUS
J2 - 38	+5V	POWER SUPPLY
J2 - 39	+15V	POWER SUPPLY
J2 - 40	-15V	POWER SUPPLY
J2 - 41	+10V	POWER SUPPLY

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TABLE IV. Test connector. - Continued

CONNECTOR PIN	SIGNAL NAME	
J2 - 42	-10V	POWER SUPPLY
J2 - 43	WDINH*	WATCH DOG INHIBIT
J2 - 44	ST01	MASTER CLEAR
J2 - 45	ST02	MICROCONTROLLER
J2 - 46	ST03	REFRESH IMPLEMENT
J2 - 47	ST04	
J2 - 48	ST05	#1 APU READY
J2 - 49	ST06	#1 APU ENDED CALCULATION
J2 - 50	ST07	A/D READY
J2 - 51	ST08	#2 APU READY
J2 - 52	ST09	#2 APU ENDED CALCULATION
J2 - 53	ST10	RECEIVE BUFFER FULL (UART)
J2 - 54	ST11	
J2 - 55	ST12	
J2 - 56	ST13	
J2 - 57	ST14	
J2 - 58	+20V	POWER SUPPLY
J2 - 59	-20V	POWER SUPPLY
J2 - 60	3V RMS	400 Hz REFERENCE
J2 - 61	SP	SAMPLE PULSE

* Represents an Active Low Signal

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TABLE IV. Test connector. - Continued

CONNECTOR PIN	SIGNAL NAME	
J2 - 62	EXTOSC	EXTERNAL OSCILLATOR INPUT
J2 - 63	OSCEN	OSCILLATOR ENABLE
J2 - 64	IN00*	#1 APU PREP LOW BITS
J2 - 65	IN01*	#1 APU PREP HIGH BITS
J2 - 66	IN02*	#1 APU RESET/READ DATA
J2 - 67	IN03*	#2 APU PREP LOW BITS
J2 - 68	IN04*	#2 APU PREP HIGH BITS
J2 - 69	IN05*	#2 APU RESET/READ DATA
J2 - 70	IN06*	RESET BIT TIMER
J2 - 71	IN07*	BALLISTICS DATA
J2 - 72	IN08*	MASTER CLEAR STATUS RESET
J2 - 73	IN09*	SDC INTERFACE DATA
J2 - 74	IN10*	SDC TEST ENABLE
J2 - 75	IN11*	
J2 - 76	IN12*	
J2 - 77	IN13*	
J2 - 78	IN14*	
J2 - 79	IN15*	A/D DATA RESET
J2 - 80	MWR*	MEMORY WRITE
J2 - 81	MOE*	MEMORY ENABLE

* Represents an Active Low Signal

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TABLE IV. Test connector. - Continued

CONNECTOR PIN	SIGNAL NAME	
J2 - 82	CTRL	CONTROL
J2 - 83	HALT*	PROCESSOR HALT
J2 - 84	DDLD	DISPLAY DATA LOAD
J2 - 85	OUT00*	OUTPUT APU#1 LSB
J2 - 86	OUT01*	OUPUT APU#1 MSB
J2 - 87	OUT02*	APU #1 SET COMMAND
J2 - 88	OUT03*	OUTPUT APU#2 LSB
J2 - 89	OUT04*	OUTPUT APU#2 MSB
J2 - 90	OUT05*	APU#2 SET COMMAND
J2 - 91	OUT06*	SIN/COS ENABLE
J2 - 92	OUT07*	SIN/COS DISABLE
J2 - 93	OUT08*	BIT FLAG (INDICATOR)
J2 - 94	OUT09*	REFRESH IMPLEMENT OFF
J2 - 95	OUT10*	OUPUT SDC INTERFACE DATA
J2 - 96	OUT11*	SDC INTERFACE TEST DISABLE
J2 - 97	OUT12*	
J2 - 98	OUT13*	
J2 - 99	OUT14*	
J2 -100	OUT15*	OUTPUT A/D MULTIPLEXER
J2 -101	BIRCK*	BUFFERED INSTR. REGISTER CLOCK

* Represents an Active Low Signal

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TABLE IV. Test connector. - Continued

CONNECTOR PIN	SIGNAL NAME	
J2 -102	INTRPT	INTERRRUPT
J2 -103	A11E1-DG	DIGITAL GROUND
J2 -104	A11E2-SG	SIGNAL GROUND

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3.5.2 Input/output signals. The input interface section and the output section shall provide the electrical and functional interface between the Weapon Aiming Computer and the connected associated equipments. Table V describes the input/output signals and the intended associated equipment usage that shall be provided to the WAC Unit.

3.5.3 Insulation resistance. The insulation resistance shall be not less than 20 megohms at 250 VDC when measured between the points listed below:

	Connector Pin Low	Connector Pin High
a.	JIB-32	J1B-35
b.	J1B-33	J1B-35
c.	J1B-34	J1B-35
d.	J1B-32	J1B-33
e.	J1B-32	J1B-34
f.	J1B-33	J1B-34
g.	Chassis	J1B-32
h.	Chassis	J1B-33
i.	Chassis	J1B-34
j.	Chassis	J1B-35

3.5.4 Bonding. The resistance between the case bonding point and connector ground pins shall be not greater than 0.50 ohms when measured between the points listed below:

	Connector Pin Low	Connector Pin High
a.	Chassis	J1A-1
b.	Chassis	J1A-3
c.	Chassis	J1B-3

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TABLE V. Input/output signals.

DC analog input voltage

Signal	Source	Connector Pin	Nominal Input Voltage	Sensitivity	Signal Range	Remarks
Target Height	Camera Control Panel	J1A-30	0 to +10 V	1000 ft/volt	0 to +10,000 ft	
Stadiametric Range	Throttle Twistgrip	J1A-31	0 to +10 V	600 ft/volt	0 to 6000 ft	
Indicated Airspeed	Air Data Computer	J1A-62	0 to +10 V	.239 VDC/10kts 0Vdc=250kts	650 to 660 kts	
True Airspeed	Air Data Computer	J1A-63	0 to +10 V	.1589VDC/10kts .19Vdc=200kts	200 to 750 kts	
Barometric Height	Air Data Computer	J1A-64	0 to +10 V	.1281VDC/1000ft .425VDC=1000ft	-1000 to 60,000 ft	
Barometric Datum Pressure Setting	Signal Data Converter	J1A-65	0 to +10 V	1300 ft/volt	-6,286 to +1,396	
Angle of Attack	Angle of Attack Sensor	J1A-66	0 to +10 V	.33VDC/deg. 1.666VDC=0°	-8.334° to 41.666°	
Mach Number	Signal Data Converter	J1A-50	0 to +10 V	.8781VDC/.1M .14VDC=.3M	.3 to 1.4 M	
Stick Interval	Weapons Control Panel	J1A-10	0 to +10 V	.0352VDC/sec	20, 40, 80, 160 360 ms	4.3 K Ohm Internal Load
Stick Length	Weapon Control Panel	J1A-11	0 to +10 V	-.368VDC/sec	20 to 3600 ms	15 K Ohm Internal Load

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TABLE V. Input/output signals. - Continued

Discrete Input Signals

Signal	Source	Connector Pin	Nominal Input Voltage	Sensitivity	Signal Range	Remarks
Air-to-Air "ON"	Signal Data Converter	J1A-9	28 VDC	28 V ± 1V	ON 28V OFF 0V	
Weapons B "ON"	Display Set Control	J1A-21	13.3 VDC	±.3V	ON 0V OFF 13.3V	OFF Voltage Value Per Pullup Resistor in SDC
Store Heading	Control Column	J1A-51	28 VDC	28 V ± 1V	ON 28V OFF 0V	
Weight On Wheels (Store Baro Height)	Weight On Wheels Microswitch	J1B-51	28 VDC	28 V ± 1V	ON 28VDC OFF 0V	Switch Located on Oleo Strut
Pilots Confidence Test	Center Console Switch	J1B-8	0V		ON ground OFF open	Off Internally pulled up to +10V
Reset Implement	Signal Data Converter	J1A-8	DC Pulse 5V DC	Standard T^2L Signal	0	Freq. 40.69 Hz pulse

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TABLE V. Input/output signals. - Continued

AC analog input voltage

Signal	Source	Connector Pin	Nominal Input Voltage	Sensitivity	Signal Range	Remarks
Pitch Rate	Twin-Axis Rate Sensor	J1B-41	± 4 VRMS	200 mv/deg/sec	± 20 deg/sec	
YAW Rate	Twin-Axis Rate Sensor	J1B-39	± 4 VRMS	200 mv/deg/sec	± 20 deg/sec	
Roll Rate	Single-Axis Rate Sensor	J1B-37	±1.8 VRMS	45 mv/deg/sec	± 40 deg/sec	

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TABLE V. Input/output signals. - Continued

3 Wire Synchro Input Signals

Signal	Source	Connector Pin	Nominal Input Voltage	Sensitivity	Signal Range	Remarks
Pitch Angle	Electronic Central Amplifier	J1B-21 Y (S3) J1B-22 X (S1) J1B-23 Z (S2)	11.8 VRMS @ 400 Hz		± 90°	
Roll Angle	Electronic Central	J1B-24 X (S1) J1B-25 Y (S3) J1B-26 Z (S2)	11.8 VRMS @ 400 Hz		0 to 360°	
Heading Angle	Electronic Central Amplifier	J1B-27 Y (S3) J1B-28 X (S1) J1B-29 Z (S2)	11.8 VRMS @ 400 Hz		0 to 360°	
Tacan Bearing	Tacan Converter	J1B-18 X (S1) J1B-19 Y (S3) J1B-20 Z (S2)	11.8 VRMS @ 400 Hz		0 to 360°	

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TABLE V. Input/output signals. - Continued

Digital Input Signals

Signal	Source	Connector Pin	Nominal Input Voltage	Sensitivity	Signal Range	Remarks
Ballistics Code	Ballistics Selector Panel	J1A-12 J1A-13 J1A-14 J1A-15 J1A-16 J1A-17	0 to 5 VDC		TRUE 5V FALSE 0	J1A-12 Parity J1A-13 2 J1A-14 2 J1A-15 2 J1A-16 2 J1A-17 2
Barometric Height (Uncorrected) (Unscaled)	Signal Data Converter	J1A-6 J1A-7 J1A-11 J1A-12	Differential ±5V		-1000 to +60,000 ft	Digital Serial Transmitted on UART line
Barometric Height (Corrected)	Signal Data Converter	J1A-6 J1A-2 J1A-11 J1A-12	Differential ±5V		-1000 to +60,000 ft	Digital Signal Transmitted on UART line
Radio HT WPNS 'A' Select Radio Valid Baro or Radio Select	Signal Data Converter	J1A-6 J1A-2 J1A-11 J1A-12	Differential ±5V		0 to 5000 ft 1 = WPNS A 0 = Not WPNS A 1 = Not Radio Valid 0 = Radio Valid 1 = Barro Select 0 = Barro Select	Digital Signal Transmitted on UART line

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TABLE V. Input/output signals. - Continued

DC Reference Voltage Output Signals

Signal	Source	Connector Pin	Nominal Output Voltage	Accuracy	Current Limiting	Remarks
Ballistics Selector Panel Ref. Voltage	WAC	J1A-36 J1A-37	+ 5VDC	±50mv	65ma	J1A-37 0 Volt
Weapon Control Panel Ref. Voltage	WAC	J1A-38 J1A-39	+ 10VDC	±50mv	30ma	J1A-39 0 Volt
Weapon Control Panel Ref. Voltage	WAC	J1A-40 J1A-39	- 10VDC	±50mv	30ma	J1A-39 0 Volt
Vertical Speed Transducer Ref. Voltage	WAC	J1A-52 J1A-53	+ 15VDC	±50mv	65ma	J1A-53 0 Volt
Vertical Speed Transducer Ref. Voltage	WAC	J1A-54 J1A-53	- 15VDC	±50mv	65ma	J1A-53 0 Volt
Air Data Converter	WAC	J1A-35 J1A-34	+ 10VDC	±50mv	50ma	J1A-34 0 Volt
Target Height Set Potentiometer	WAC	J1B- 1 J1A- 2	+ 10VDC	±50mv	50ma	J1A-2 0 Volt

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TABLE V. Input/output signals. - Continued

Discrete Output Signals

Signal	Destination	Connector Pin	Nominal Output Voltage	Sensitivity	Signal Range	Remarks
WAC Fail	To Centralized Warning Panel	J18-14	28Vdc	-	0 or 28Vdc	

DC Current Input

Signal	Source	Connector Pin	Nominal Input Current Swing	Sensitivity	Signal Range	Remarks
UHF Homer	UHF Homer Adapter	J1A-32 J1A-33	± 150uA	5uA/degree	± 30°	

DC Current Output

Signal	Source	Connector Pin	Nominal Output Current Swing	Sensitivity	Signal Range	Remarks
UHF Homer	ADI	J1A-28 J1A-29	± 2.385mA	79.5uA/degree	± 30°	into a 1000 ohm load

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TABLE V. Input/output signals. - Continued

Digital Output Signals

Signal	Range	Scale	Remarks
Bank Angle (COS)	Cos (0 to 360°) +1 to -1	+1 = 32767 0 = 0 -1 = -32767	
Bank Angle (SIN)	Sin (0 to 360°) +1 to -1	+1 = 32767 0 = 0 -1 = -32767	
Heading (Angle)	0 to 180°	180°= 0 90°= 16387 0°= 32767	
Heading Sector	0 to -1	0° 0 = 180° -1 = 180° 360°	
Elevation (Pitch Angle)	+90°to-90°	+90°= 32767 0°= 0 -90°= -32767	
Tacan (Angle)	-16°to 0°to -16°	-16°=-32767 0°= 0 +16°= 32767	
True Air Speed	0 to 673.18 Kts	0Kts=-1048 673.18Kts=32767	
Aiming Mark (Elevation)	+16° to -16°	+16°= 32767 0°= 0 -16°=-32767	Referenced to Tube Center

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TABLE V. Input/output signals. - Continued

Digital Output Signals

Signal	Range	Scale	Remarks
Aiming Mark (Azimuth)	+16° to -16°	+16° = 32767 0° = 0 -16° =-32767	
Ballistics Not Valid	Valid or Not Valid	0 = Valid -1 = Not Valid	
CCIP Cue (Elevation)	+2° to -18°	+2°= 3641 0°= 0 -18°=-32767	Referenced from LFD Count Value limited internally
CCIP Cue (Azimuth)	+10° to -10°	+10°= 6143 0°= 0 -10°=-6143	Count Value limited internally
Azimuth Steering Line (Angle)	+180° to -180°	180° = 32767 0° = 0 -180°=-32767	Count Value limited internally
Azimuth Steering Line (Length)	o to 14.14°	0° = 0 14.14° = 2263	Count Value limited internally
Reflection Length	0 to 14.14°	0° = 0 14.14° = 2263	MSB=0 Display Solid Line MSB=1 Display Dashed Line

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3.5.5 Continuity. The resistance measured between the pins listed below shall be not greater than 1.0 ohms.

	Connector/Pin		
a.	J1 shell	and	J2 shell
b.	J1A-1	and	J2 shell
c.	J1A-34	and	J1A-5
d.	J1A-34	and	J1A-37
e.	J1A-34	and	J1A-39
f.	J1A-34	and	J1A-53
g.	J1A-34	and	J1B-2
h.	J1A-4	and	J1A-2
i.	J1A-4	and	J1A-7
j.	J1A-21	and	J1A-22

3.5.6 Power supply. The computer shall be capable of producing all DC voltages required to operate the computer. Primary power service shall be as specified in 3.3.13. The power supply voltages shall be as specified (throughout) below:

	Connector/Pin High	Low	Voltage	Tolerance
a.	J2-38	J2-103	+5 V	± .2V DC
b.	J2-39	J2-104	+15 V	±.25V DC
c.	J2-40	J2-104	-15 V	±.25V DC
d.	J2-41	J2-104	+10 V	± .1V DC
e.	J2-42	J2-104	-10 V	± .1V DC
f.	J2-58	J2-104	+20 V	±2.0V DC
g.	J2-59	J2-104	-20 V	±2.0V DC

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3.5.7 Syncro input signals.

3.5.7.1 Elevation angle (pitch). The computer shall receive at Connector/Pins J1B-21 (y), J1B-22 (x), J1B-23 (z), synchro signals representing elevation angle (pitch) and converting this signal within the unit to a serial digital outputs. The outputs shall be as indicated below:

	Input	Output
a.	0°	0° ± 0.5°
b.	30°	30° ± 0.8°
c.	60°	60° ± 1.1°
d.	90°	90° ± 1.4°
e.	-90°	-90° ± 1.4°
f.	-60°	-60° ± 1.1°
g.	-30°	-30° ± 0.8°

3.5.7.2 Bank angle (roll). The computer shall receive at Connector/Pins J1B-24 (x), J1B-25 (y), J1B-26 (z), 3 synchro signals representing bank angle and converting the signals within the unit to a serial digital outputs. The outputs shall be as indicated below:

	Input	Bank Angle (Sin)	Bank Angle (Cos)
a.	0°	0 ± .01	1 ± .01
b.	30°	.5 ± .01	.866 ± .01
c.	60°	.866 ± .01	.5 ± .01
d.	90°	1.0 ± .01	0 ± .01
e.	120°	.866 ± .01	-.5 ± .01
f.	150°	.5 ± .01	-.866 ± .01
g.	180°	0 ± .01	-1 ± .01
h.	210°	-.5 ± .01	-.866 ± .01
i.	240°	-.866 ± .01	-.5 ± .01
j.	270°	-1 .01	0 ± .01
k.	300°	-.866 .01	.5 ± .01
l.	330°	-.5 ± .01	.866 ± .01

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3.5.7.3 Heading angle. The computer shall receive at Connector/Pins J1B-27 (y), J1B-28 (x), J1B-29 (z), synchro signals representing heading angle and converting the signals within the unit to a serial digital outputs. The output shall be as indicated below:

	Input	Output
a.	0°	0° ± 0.8°
b.	30°	30° ± 0.8°
c.	60°	60° ± 0.8°
d.	90°	90° ± 0.8°
e.	120°	120° ± 0.8°
f.	150°	150° ± 0.8°
g.	180°	180° ± 0.8°
h.	200°	200° ± 0.8°
i.	240°	240° ± 0.8°
j.	280°	280° ± 0.8°
k.	300°	300° ± 0.8°
l.	330°	330° ± 0.8°

3.5.7.4 Tacan angle. The computer shall receive at Connector/Pins J1B-18 (x), J1B-19 (y), J1B-20 (z), synchro signals representing tacan angle and converting this signal within the unit to a serial digital output. The outputs shall be as indicated below:

	Input	Output
a.	0°	0 ± .014
b.	1°	.0174 ± .015
c.	359°	-.0174 ± .015
d.	5°	.087 ± .016
e.	355°	-.087 ± .016
f.	10°	.173 ± .018
g.	350°	-.173 ± .018
h.	15°	.258 ± .019
i.	345°	-.258 ± .019
j.	185°	-.087 ± .001
k.	175°	.087 ± .001

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3.5.8 True airspeed (TAS). The computer shall receive at Connector/Pin J1A-63, a DC analog voltage representing TAS, converting this signal within the unit to a serial digital output. The output shall be as indicated below:

	Input	Output
a.	200 kts	200 ± 3 kts
b.	250	250 ± 3 kts
c.	300	300 ± 3 kts
d.	350	350 ± 3 kts
e.	400	400 ± 3 kts
f.	500	500 ± 3 kts
g.	550	550 ± 3 kts
h.	600	600 ± 3 kts
i.	650	650 ± 3 kts

3.5.9 U.H.F. homer. The computer shall receive at Connector/Pins J1A-32 (+ve), J1A-33 (-ve), DC Voltage representing UHF Homer steering demands, the signal is buffered within the unit with the output on connector/pins J1A-28 (+ve), J1A-29 (-ve) and the outputs shall be as listed below:

	Input	Output
a.	7 VDC	7 ± .049 VDC
b.	7 VDC	7 ± .049 VDC
c.	0 VDC	0 ± .049 VDC
d.	0 VDC	-10 ± 2 VDC

3.5.10 Air-to-air. The computer shall receive from associated equipment input signals that shall enable the computer to compute, air-to-air target range, lead angle computation for gun aiming, the information shall be transmitted to the Signal Data Converter which will produce symbols for display on the Digital Display Indicator. Table VI outlines the input signals for thirteen (13) air combat maneuvers and the required output signals and tolerances (Referenced to CRT tube Center).

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TABLE VI Air-to-Air test.

	INPUTS										OUTPUTS		
Test No.	Air-to-Air	Range	TAS	AOA	Pitch Rate	Yaw Rate	Roll Rate	Roll Angle	Pitch Angle	Baro. Ht.	Az.	El.	Range
1	ON	2000	350	2°	0	0	0	0	0	3000	0	7.553	2000
2	ON	2000	350	0°	0	0	0	0	90	3000	0	7.871	2000
3	ON	2000	350	-2°	0	0	0	270	0	3000	.327	7.484	2000
4	ON	600	350	-3°	+10°/s	0	0	0	0	3000	0	4.981	600
5	ON	600	350	3°	-10°/s	0	0	0	0	3000	0	9.999	600
6	ON	1200	400	-3°	+10°/s	0	0	0	0	18000	0	2.395	1200
7	ON	1200	400	3°	-10°/s	0	0	0	0	18000	0	12.825	1200
8	ON	1200	400	2°	0	+10°/s	0	0	0	18000	-5.113	7.678	1200
9	ON	1200	400	2°	0	-10°/s	0	0	0	18000	5.113	7.678	1200
10	ON	600	350	-2°	+7.5°/s	0	+33°/s	330	0	3000	1.418	6.122	600
11	ON	600	350	2°	0	+7.5°/s	+33°/s	330	0	3000	-1.184	6.501	600
12	ON	1200	400	2°	0	+5°/s	+22°/s	30	0	9000	-2.497	8.563	1200
13	ON	1200	400	-1°	+ 5°/s	0	+22°/s	30	0	9000	-.87	5.075	1200

Tolerance - Azimuth ±.25°
Elevation ±.25°
Range ± 100'

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3.5.11 Air-to-surface. The computer shall receive from associated equipment input signals that shall enable the computer to compute; Advance Breakaway Warning, Breakaway Warning, Air-to-Ground Target Height, Ballistics Valid or Not Valid, Continuous Computed Inpact Point Azimuth and Elevation, Azimuth Steering Line Angle and Length, and Continuous Computed Impact Point Reflected Length signals. The information shall be transmitted to the Signal Data Converter which will produce symbols for display on the Digital Display Indicator. Table VII outlines the input signals for two indivdual runs with the required output signals and tolerances for the six weapons listed. Table VIII, IX, X and XI deal similarly with the remaining weapons (Reference Table II).

3.5.12 Radio altimeter. The computer shall receive from associated equipment input signals and Radio Height information enabling the computer to compute weapon delivery release conditions. This information shall be transmitted to the SDC for display on the DDI. Table XIV outlines the input signals and the required output signals with tolerance are specified in Table XIV.

3.5.13 Weapon 'A'mode. The computer shall receive from associated equipment input signal enabling the computer to compute weapon deliver release condition for Weapons A mode (no wind solution). The information shall be transmitted to the SDC for display on the DDI. Table XV outlines the input signals and the required output signals with tolerance are specified in Table XIV.

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TABLE VII. Air-to-Surface test.

For Weapons 1, 2, 3, 4, 5, 6 (Reference Table 2)

INPUT SIGNALS		RUN #1	RUN #2
PITCH RATE		5°/s/ 1VRMS	2.5°/s/.5VRMS
ANGLE OF ELEVATION		-60°/-60°	-45°/-45°
ANGLE OF BANK		0°/0°	0°/0°
HEADING		0°/0°	0°/0°
BAROMETRIC HEIGHT		10,000ft/1.834VDC	10,000ft/1.834VDC
PRESSURE DATUM		200ft/5.971VDC	-200ft/6.278VDC
TARGET HEIGHT		1,200ft/1.2VDC	4,800ft/4.8VDC
BAROMETRIC HT. (AIRFIELD)		1,000ft/.6812VDC	500ft/.61715VDC
HUD BAROMETRIC HT.	(CORRECTED FOR PRESSURE DATUM)	800ft/.65058VDC	700ft/.64277VDC
INDICATED AIRSPEED		426kts/4.22VDC	497kts/5.92VDC
TRUE AIRSPEED		444kts/4.073VDC	532kts/5.48VDC
MACH NUMBER		.7/3.652VDC	.9/5.409VDC
ANGLE OF ATTACK		1.73°/2.243VDC	2.87°/2.625VDC
STICK INTERVAL		.16/4.478VDC	.02/.56VDC
STICK LENGTH		3/-1.091VDC	1/-.045VDC
HEADING		5°/-5°	5°/-5°
	RADIO HEIGHT	0	0
	BARO OR RADIO SELECT	BARO	BARO
RADAR	RADIO HT VALID OR INVALID	VALID	VALID
	WEAPON ON OR OFF	OFF	OFF

Output Signal	Tolerance
A = CCIP AZIMUTH	±.6°
B = CCIP ELEVATION	±.3°
L = CCIP ASL ANGLE	± 5°
L_2= CCIP ASL LENGTH	±.83°
L_1= CCIP REFLECTION LENGTH	±.83°
L_3= CCIP BOUNDARY DISCRIMINANT	
B_9= CCIP BREAKAWAY DISCRIMINANT	
B_5= CCIP ADVANCED BREAKAWAY WARNING LENGTH	.5°
S_9= CCIP ADVANCED BREAKAWAY DISCRETE	-
T_1= CCIP TARGET HEIGHT	±50

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TABLE VII. Air-to-Surface test. - Continued

RUN #1

OUTPUT SIGNALS	ZERO WIND	TARGET TRACKING
#1		
A	0	-3.4
B	-7.05	-7.05
L	0	3.99
L2	0	0
L1	5.32	5.33
L3	0	0
B9	0	0
B5	0	2.666
S9	0	1
T1	0	1200
#2		
A	0	-3.46
B	-7.22	-7.22
L	0	1.36
L2	0	0
L1	5.45	5.495
L3	0	0
B9	0	0
B5	0	2.666
S9	0	1
T1	0	1200
#3		
A	0	-3.43
B	-7.37	-7.37
L	0	2.31
L2	0	0
L1	5.64	5.64
L3	0	0
B9	0	0
B5	0	2.666
S9	0	1
T1	0	1200
#4		
A	0	-3.47
B	-7.03	-7.03
L	0	.92
L2	0	0
L1	5.30	5.30
L3	0	0
B9	0	0
B5	0	2.666
S9	0	1
T1	0	1200
#5		
A	0	-3.45
B	-7.17	-7.17
L	0	1.81
L2	0	0
L1	5.44	5.44
L3	0	0
B9	0	0
B5	0	2.666
S9	0	1
T1	0	1200
#6		
A	0	-3.48
B	-6.9	-6.87
L	0	.592
L2	0	0
L1	5.16	5.16
L3	0	0
B9	0	0
B5	0	2.666
S9	0	1
T1	0	1200

RUN #2

OUTPUT SIGNALS	ZERO WIND	TARGET TRACKING
#1		
A	0	-3.27
B	-6.17	-6.17
L	0	3.58
L2	0	0
L1	3.3	3.3
L3	0	0
B9	-1	-1
B5	0	0
S9	0	4800
T1	0	0
#2		
A	0	-3.41
B	-6.59	-6.59
L	0	.0595
L2	0	0
L1	3.72	3.72
L3	0	0
B9	-1	-1
B5	0	0
S9	0	4800
T1	0	0
#3		
A	0	-3.34
B	-6.59	-6.59
L	0	-.14
L2	0	0
L1	3.72	3.72
L3	0	0
B9	-1	-1
B5	0	0
S9	0	4800
T1	0	0
#4		
A	0	-3.43
B	-6.4	-6.4
L	0	-.15
L2	0	0
L1	3.53	3.53
L3	0	0
B9	-1	-1
B5	0	0
S9	0	4800
T1	0	0
#5		
A	0	-3.38
B	-6.45	-6.45
L	0	-.18
L2	0	0
L1	3.58	3.58
L3	0	0
B9	0	0
B5	0	0
S9	0	4800
T1	0	0
#6		
A	0	-3.46
B	-6.3	-6.3
L	0	.063
L2	0	0
L1	3.43	3.43
L3	0	0
B9	-1	-1
B5	0	0
S9	0	4800
T1	0	0

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TABLE VIII.

For Weapons 7, 8, 9, 10 (Reference Table 2)

INPUT SIGNALS		RUN #1	RUN #2
PITCH RATE		2.5°/s/.5VRMS	2.5°/s/.5VRMS
ANGLE OF ELEVATION		-25°/-25°	-15°/-15°
ANGLE OF BANK		0°/0°	0°/0°
HEADING		0°/0°	0°/0°
BAROMETRIC HEIGHT		4,000ft/1.066VDC	6,900ft/1.437VDC
PRESSURE DATUM		0ft/6.125VDC	400ft/5.817VDC
TARGET HEIGHT		2,470ft/2.47VDC	4,800ft/4.8VDC
BAROMETRIC HT. (AIRFIELD)		2,000ft/.8093VDC	2,400ft/.86054VDC
HUD BAROMETRIC HT. (CORRECTED FOR PRESSURE DATUM)		2,000ft/.8093VDC	2,000ft/.8093VDC
INDICATED AIRSPEED		337.7kts/2.096VDC	426kts/4.22VDC
TRUE AIRSPEED		343.6kts/2.472VDC	444kts/4.073VDC
MACH NUMBER		.55/2.335VDC	.7/3.652VDC
ANGLE OF ATTACK		2.498°/2.499VDC	2.87°/2.625VDC
STICK INTERVAL		.32/8.956VDC	.08/2.239VDC
STICK LENGTH		1/-.727VDC	6/-.1.091VDC
HEADING		355°/-5°	355°/-5°
	RADIO HEIGHT	0	0
	BARO OR RADIO SELECT	BARO	BARO
RADAR	RADIO HT VALID OR INVALID	VALID	VALID
	WEAPON ON OR OFF	OFF	OFF

Output Signal	Tolerance
A = CCIP AZIMUTH	±.6°
B = CCIP ELEVATION	±.3°
L = CCIP ASL ANGLE	± 5°
L_2= CCIP ASL LENGTH	±.83°
L_1= CCIP REFLECTION LENGTH	±.83°
L_3= CCIP BOUNDARY DISCRIMINANT	
B_9= CCIP BREAKAWAY DISCRIMINANT	
B_5= CCIP ADVANCED BREAKAWAY WARNING LENGTH	.5°
S_9= CCIP ADVANCED BREAKAWAY DISCRETE	-
T_1= CCIP TARGET HEIGHT	±50

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TABLE VIII. - Continued

RUN #1

OUTPUT SIGNALS	ZERO WIND	TARGET TRACKING
#1		
A	0	4.07
B	-13.73	-13.73
L	0	-9.2
L2	0	0
L1	11.23	11.38
L3	0	0
B9	-1	-1
B5	0	0
S9	0	0
T1	2470	2470
#2		
A	0	3.1
B	-8.85	-8.85
L	0	-1.9
L2	0	0
L1	6.35	6.35
L3	0	0
B9	-1	-1
B5	0	0
S9	0	0
T1	2470	2470
#3		
A	0	3.1
B	-11.55	-11.55
L	0	---
L2	0	0
L1	2.05	6.35
L3	0	0
B9	-1	-1
B5	0	0
S9	0	0
T1	2470	2470
#4		
A	0	3.17
B	-10.05	-10.05
L	0	-5.84
L2	0	0
L1	7.5	9.1
L3	0	0
B9	-1	-1
B5	0	0
S9	0	0
T1	2470	2470

RUN #2

OUTPUT SIGNALS	ZERO WIND	TARGET TRACKING
#1		
A	0	5.82
B	-17.22	-17.22
L	0	-13.98
L2	0	0
L1	14.25	14.79
L3	0	0
B9	0	0
B5	1.339	1.339
S9	0	0
T1	4800	4800
#2		
A	0	3.2
B	-9.58	-9.58
L	0	-4.93
L2	0	0
L1	6.71	6.73
L3	0	0
B9	0	0
B5	1.339	1.339
S9	0	0
T1	4800	4800
#3		
A	0	4.63
B	-14.23	-14.17
L	0	-11.91
L2	0	0
L1	11.36	11.55
L3	0	0
B9	0	0
B5	1.339	1.339
S9	0	0
T1	4800	4800
#4		
A	0	3.81
B	-12.06	-12.06
L	0	-9.65
L2	0	0
L1	9.19	9.32
L3	0	0
B9	0	0
B5	1.339	1.339
S9	0	0
T1	4800	4800

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TABLE IX.

For Weapons 11, 12, 13, 14 (Reference Table 2)

INPUT SIGNALS		RUN #1	RUN #2
PITCH RATE		-2.5°s-.5VRMS	
ANGLE OF ELEVATION		-15°/-15°	
ANGLE OF BANK		0°/0°	
HEADING		0°/0°	
BAROMETRIC HEIGHT		6,900ft/1.437VDC	
PRESSURE DATUM		400ft/5.817VDC	
TARGET HEIGHT		4,800ft/4.8VDC	
BAROMETRIC HT. (AIRFIELD)		2,400ft/.86054VDC	
HUD BAROMETRIC HT. (CORRECTED FOR PRESSURE DATUM)		2,000ft/.8093VDC	
INDICATED AIRSPEED		426kts/4.22VDC	
TRUE AIRSPEED		444kts/4.073VDC	
MACH NUMBER		.7/3.652VDC	
ANGLE OF ATTACK		2.87°/2.625VDC	
STICK INTERVAL		.08/2.239VDC	
STICK LENGTH		6/-1.091VDC	
HEADING		-5°/-5°	
	RADIO HEIGHT	0	
	BARO OR RADIO SELECT	BARO	
RADAR	RADIO HT VALID OR INVALID	VALID	
	WEAPON ON OR OFF	OFF	

Output Signal	Tolerance
A = CCIP AZIMUTH	±.6°
B = CCIP ELEVATION	±.3°
L = CCIP ASL ANGLE	± 5°
L_2= CCIP ASL LENGTH	±.83°
L_1= CCIP REFLECTION LENGTH	±.83°
L_3= CCIP BOUNDARY DISCRIMINANT	
B_9= CCIP BREAKAWAY DISCRIMINANT	
B_5= CCIP ADVANCED BREAKAWAY WARNING LENGTH	.5°
S_9= CCIP ADVANCED BREAKAWAY DISCRETE	-
T_1= CCIP TARGET HEIGHT	±50

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TABLE IX. - Continued

RUN #1

OUTPUT SIGNALS	ZERO WIND	TARGET TRACKING
#1		
A	0	3.05
B	-9.83	-9.83
L	0	-6.59
L2	0	0
L1	6.96	7
L3	0	0
B9	0	0
B5	1.339	1.339
S9	0	0
T1	4800	4800
#2		
A	0	.45
B	-8.09	-8.09
L	0	25.77
L2	0	0
L1	5.22	5.88
L3	0	0
B9	0	0
B5	1.339	1.339
S9	0	0
T1	4800	4800
#3		
A	0	2.97
B	-9.58	-9.58
L	0	-6.18
L2	0	0
L1	6.71	6.75
L3	0	0
B9	0	0
B5	1.339	1.339
S9	0	0
T1	4800	4800
#4		
A	0	1.62
B	-7.99	-7.99
L	0	42.81
L2	0	0
L1	5.12	6.98
L3	0	0
B9	0	0
B5	1.339	1.339
S9	0	0
T1	4800	4800

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TABLE X.

For Weapons 15, 16, 17, 18, 19, 20 (Reference Table 2)

INPUT SIGNALS		RUN #1	RUN #2
PITCH RATE		5°/s /1VRMS	0°/s0VRMS
ANGLE OF ELEVATION		-60°/-60°	-75°/-75°
ANGLE OF BANK		0°/0°	0°/0°
HEADING		0°/0°	0°/0°
BAROMETRIC HEIGHT		10,000ft/1.834VDC	8,100ft/1.591VDC
PRESSURE DATUM		200ft/5.971VDC	0ft/6.125VDC
TARGET HEIGHT		1,200ft/1.2VDC	0ft/0VDC
BAROMETRIC HT. (AIRFIELD)		1,000ft/.6812VDC	0ft/.533VDC
HUD BAROMETRIC HT. (CORRECTED FOR PRESSURE DATUM)		800ft/.65058VDC	0ft/.533VDC
INDICATED AIRSPEED		426kts/4.22VDC	480kts/5.523VDC
TRUE AIRSPEED		444kts/4.073VDC	532kts/5.48VDC
MACH NUMBER		.7/3.652VDC	.83/4.794VDC
ANGLE OF ATTACK		1.73°/2.243VDC	0°/1.667VDC
STICK INTERVAL		.16/4.478VDC	5°/5°
STICK LENGTH		3/-1.091VDC	
HEADING		5°/5°	
	RADIO HEIGHT	0	0
	BARO OR RADIO SELECT	BARO	BARO
RADAR	RADIO HT VALID OR INVALID	VALID	VALID
	WEAPON ON OR OFF	OFF	OFF

Output Signal	Tolerance
A = CCIP AZIMUTH	±.6°
B = CCIP ELEVATION	±.3°
L = CCIP ASL ANGLE	± 5°
L_2= CCIP ASL LENGTH	±.83°
L_1= CCIP REFLECTION LENGTH	±.83°
L_3= CCIP BOUNDARY DISCRIMINANT	
B_9= CCIP BREAKAWAY DISCRIMINANT	
B_5= CCIP ADVANCED BREAKAWAY WARNING LENGTH	.5°
S_9= CCIP ADVANCED BREAKAWAY DISCRETE	-
T_1= CCIP TARGET HEIGHT	±50

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TABLE X. - Continued

RUN #1

OUTPUT SIGNALS	ZERO WIND	TARGET TRACKING
#1		
A	0	-1.41
B	-2.76	-2.76
L	0	0
L2	0	0
L1	1.03	1.03
L3	0	0
B9	0	0
B5	2.666	2.666
S9	0	1
T1	1200	1200
#2		
A	-	-1.64
B	-3.0	-3.0
L	0	0
L2	0	0
L1	1.26	1.20
L3	0	0
B9	0	0
B5	2.666	2.666
S9	0	1
T1	1200	1200
#3		
A	0	-1.29
B	-2.59	-2.58
L	0	0
L2	0	0
L1	.85	.85
L3	0	0
B9	0	0
B5	2.666	2.666
S9	0	1
T1	1200	1200
#4		
A	0	-1.3
B	-2.61	-2.61
L	0	0
L2	0	0
L1	.876	.88
L3	0	0
B9	0	0
B5	2.666	2.666
S9	0	1
T1	1200	1200
#5		
A	0	-1.15
B	-2.62	-2.62
L	0	0
L2	0	0
L1	.89	.89
L3	0	0
B9		0
B5	2.666	2.666
S9	0	1
T1	1200	1200
#6		
A	0	-1.16
B	-2.65	-2.65
L	0	0
L2	0	0
L1	.911	.911
L3	0	0
B9	0	0
B5	2.666	2.666
S9	0	1
T1	1200	1200

RUN #2

OUTPUT SIGNALS	ZERO WIND	TARGET TRACKING
#1		
A	0	-1.61
B	-.739	- .739
L	0	0
L2	0	0
L1	.739	.739
L3	0	0
B9	-1	-1
B5	0	0
S9	0	0
T1	0	0
#2		
A	0	-1.85
B	-.902	-.902
L	0	0
L2	0	0
L1	.902	.902
L3	0	0
B9	-1	-1
B5	0	0
S9	0	0
T1	0	0
#3		
A	0	-1.52
B	-.778	-.778
L	0	0
L2	0	0
L1	.778	.778
L3	0	0
B9	-1	-1
B5	0	0
S9	0	0
T1	0	0
#4		
A	0	-1.52
B	-.79	-.79
L	0	0
L2	0	0
L1	.79	.79
L3	0	0
B9		-1
B5	0	0
S9	0	0
T1	0	0
#5		
A	0	-1.38
B	-.64	-.64
L	0	0
L2	0	0
L1	.64	.64
L3	0	0
B9		-1
B5	0	0
S9	0	0
T1	0	0
#6		
A	0	-1.39
B	-.649	-.649
L	0	0
L2	0	0
L1	.649	.649
L3	0	0
B9		-1
B5	0	0
S9	0	0
T1	0	0

All units are in degrees measured from the LFD

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TABLE XI.

For Weapon 21 (Reference Table 2)

INPUT SIGNALS		RUN #1	RUN #2
PITCH RATE		5°/s /1VRMS	5°/s1VRMS
ANGLE OF ELEVATION		-60°/-60°	-45°/-45°
ANGLE OF BANK		0°/0°	60°/60°
HEADING		0°/0°	0°/0°
BAROMETRIC HEIGHT		10,000ft/1.834VDC	10,000ft/1.834VDC
PRESSURE DATUM		200ft/5.971VDC	-200ft/6.278VDC
TARGET HEIGHT		1,200ft/1.2VDC	4,800/4.8VDC
BAROMETRIC HT. (AIRFIELD)		1,000ft/.6812VDC	500ft/.61715VDC
HUD BAROMETRIC HT. (CORRECTED FOR PRESSURE DATUM)		800ft/.65058VDC	700ft/.64277VDC
INDICATED AIRSPEED		426kts/4.22VDC	498kts/5.92VDC
TRUE AIRSPEED		444kts/4.073VDC	533kts/5.485VDC
MACH NUMBER		.7/3.652VDC	.9/5.409VDC
ANGLE OF ATTACK		1.73°/2.243VDC	2.87°/2.625VDC
STICK INTERVAL			
STICK LENGTH			
HEADING		5°/5°	5°/5°
	RADIO HEIGHT	0	0
	BARO OR RADIO SELECT	BARO	BARO
RADAR	RADIO HT VALID OR INVALID	VALID	VALID
	WEAPON ON OR OFF	OFF	OFF

Output Signal	Tolerance
A = CCIP AZIMUTH	±.6°
B = CCIP ELEVATION	±.3°
L = CCIP ASL ANGLE	± 5°
L_2= CCIP ASL LENGTH	±.83°
L_1= CCIP REFLECTION LENGTH	±.83°
L_3= CCIP BOUNDARY DISCRIMINANT	
B_9= CCIP BREAKAWAY DISCRIMINANT	
B_5= CCIP ADVANCED BREAKAWAY WARNING LENGTH	.5°
S_9= CCIP ADVANCED BREAKAWAY DISCRETE	-
T_1= CCIP TARGET HEIGHT	±50

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TABLE XI. - Continued

RUN #1

OUTPUT SIGNALS #1	ZERO WIND	TARGET TRACKING
A	0	-2.96
B	-3.29	-3.29
L	0	0
L2	0	0
L1	1.56	2.50
L3	0	0
B9	0	0
B5	2.666	2.666
S9	0	0
T1	1200	1200

RUN #2

OUTPUT SIGNALS #1	ZERO WIND	TARGET TRACKING
A	-.64	-2.06
B	-.923	- .923
L	-105	-180.0
L2	0	0
L1	1.92	.512
L3	0	0
B9	0	-1
B5	0	0
S9	0	0
T1	4800	4800

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TABLE XII.

For Weapon 1 (Reference Table 2)

INPUT SIGNALS		RUN #1
PITCH RATE		5°/sec
ANGLE OF ELEVATION		-60°
ANGLE OF BANK		0°
HEADING		-
BAROMETRIC HEIGHT		0 ft
PRESSURE DATUM		0 ft
TARGET HEIGHT		0 ft
BAROMETRIC HT. (AIRFIELD)		0 ft
INDICATED AIRSPEED		426kts
TRUE AIRSPEED		444kts
MACH NUMBER		.7
ANGLE OF ATTACK		1.73°
STICK INTERVAL		.16 sec
STICK LENGTH		3
HEADING		-
	RADIO HEIGHT	5000
	BARO OR RADIO SELECT	RADIO
RADAR	RADIO HT VALID OR INVALID	VALID
	WEAPON ON OR OFF	OFF

Output Signal	Tolerance
B = CCIP ELEVATION	±.3°

RUN #1

OUTPUT SIGNALS	ZERO WIND	TARGET TRACKING
#1		
B	-5.44	0

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TABLE XIII.

For Weapon 7 (Reference Table 2)

INPUT SIGNALS			RUN #1
PITCH RATE			2.5°/sec
ANGLE OF ELEVATION			-25°
ANGLE OF BANK			0°
HEADING			0°
BAROMETRIC HEIGHT			4000ft
PRESSURE DATUM			0 ft
TARGET HEIGHT			2470ft
BAROMETRIC HT. (AIRFIELD)			2000ft
HUD BAROMETRIC HT.		(CORRECTED FOR PRESSURE DATUM)	2000ft
INDICATED AIRSPEED			338kts
TRUE AIRSPEED			344kts
MACH NUMBER			.55
ANGLE OF ATTACK			2.5°
STICK INTERVAL			.02 sec
STICK LENGTH			1
HEADING			355
		RADIO HEIGHT	0
		BARO OR RADIO SELECT	BARO
RADAR		RADIO HT VALID OR INVALID	VALID
		WEAPON ON OR OFF	ON

Output Signal	Tolerance
L = CCIP ASL ANGLE	± 5°

RUN #1

OUTPUT SIGNALS #1	ZERO WIND	TARGET TRACKING
L	-	0

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TABLE XIV

For Weapons 1 (Reference Table 2)

INPUT SIGNALS		RUN #1	RUN #2
PITCH RATE		5 deg./sec.	
ANGLE OF ELEVATION		-60 deg.	
ANGLE OF BANK		0 deg.	
HEADING		0 deg.	
BAROMETRIC HEIGHT		0 ft.	
PRESSURE DATUM		0 ft.	
TARGET HEIGHT		0 ft.	
BAROMETRIC HT. (AIRFIELD)		0 ft.	
HUD BAROMETRIC HT. (CORRECTED FOR PRESSURE DATUM)		0 ft.	
INDICATED AIRSPEED		426 kts	
TRUE AIRSPEED		444 kts	
MACH NUMBER		.7	
ANGLE OF ATTACK		1.73 deg.	
STICK INTERVAL		.16 sec.	
STICK LENGTH		3	
HEADING		—	
	RADIO HEIGHT	5000 ft.	
	BARO OR RADIO SELECT	RADIO	
RADAR	RADIO HT VALID OR INVALID	VALID	
	WEAPON ON OR OFF	OFF	

Output Signal CCIP elevation -5.44° tolerance ±.3°

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TABLE XV

For Weapons 7 (Reference Table 2)

INPUT SIGNALS		RUN #1	RUN #2
PITCH RATE		2.5 deg./sec.	
ANGLE OF ELEVATION		-25 deg.	
ANGLE OF BANK		0 deg.	
HEADING		0 deg.	
BAROMETRIC HEIGHT		4000 ft.	
PRESSURE DATUM		0 ft.	
TARGET HEIGHT		2470 ft.	
BAROMETRIC HT. (AIRFIELD)		2000 ft.	
HUD BAROMETRIC HT. (CORRECTED FOR PRESSURE DATUM)		4000 ft.	
INDICATED AIRSPEED		338 kts	
TRUE AIRSPEED		344 kts	
MACH NUMBER		.55	
ANGLE OF ATTACK		2.5 deg.	
STICK INTERVAL		.020 sec.	
STICK LENGTH		1	
HEADING		355 deg.	
	RADIO HEIGHT	0	
	BARO OR RADIO SELECT	BARO	
RADAR	RADIO HT VALID OR INVALID	-	
	WEAPON ON OR OFF	ON	

Output Signal ASL Angle 0° tolerance ±1°

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3.5.14 Invalid ballistic code. The computer shall receive at connector /pins J1A-13, J1A-14, J1A-15, J1A-16, J1A-17, J1A-12; ballistics data representing weapon selected with parity. The input/output tolerance shall be as indicated below:

Ballistic code	Parity	Output
a. Weapon 7	odd (1)	-1 Test passed
b. Weapon 7	even (0)	0 Test passed

3.5.15 Pilot's confidence test (PCC). The computer shall receive at connector/pin J1B-8, PCC signal, which will enable fixed voltages to be inputed to the circuitry of the WAC for air-to-ground and air-to-air pilot's confidence test check. The input/output tolerance shall be as indicated below:

	Air-to-surface	Engineering units
a.	Target Height	1,900 ft.
b.	nts	0
c.	ts	0
d.	TAS	451.95 kts.
e.	Baro Height	5064.95 ft.
f.	Baro Datum	-169.33 ft.
g.	IAS	406.9 kts.
h.	Mach No.	.6947796
i.	Angle of Attack	.973933 Deg.
j.	Pitch Rate	4.544996 Deg/Sec.
k.	SIN Pitch	-30 Deg.
l.	COS Pitch	-30 Deg.
m.	SIN Roll	0 Deg.
n.	COS Roll	0 Deg.

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PCC test results

a.	CCIP Azimuth	0	±.6°
b.	CCIP Elevation	-7.1°	±.6° 124±5 Depression Angle (m radians)
c.	ASL angle	0	±.42°
d.	ASL length	6.1°	±.25°
e.	TAS	452	± 3 kts.
f.	Pitch	-30°	± .8°
g.	Bank	0°	± .6°

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2). Air-to-air — Engineering units

a.	1/Range	2123.8938 ft.
b.	TAS	406.9 kts.
c.	Angle of Attack	.9739339 deg.
d.	Pitch Rate	4.544996 deg/sec.
e.	Roll Rate	0 deg/sec.
f.	Yaw Rat	0 deg/sec.
g.	Pitch Sin	-30 deg.
h.	Pitch Cos	-30 deg.
i.	Roll Sin	0 deg.
j.	Roll Cos	0 deg.

PCC test results

a.	A/A Azimuth	0°	±.25°	
b.	A/A Elevation	6.0057°	±.25°	(will coincide with -60° Pitch symbol)
c.	A/A Target Range	2100 ft.	± 100 ft.	

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3.5.16 Voltage reference test. The computer shall provide DC reference voltages required to operate associated equipment. The reference voltages shall be as indicated below:

	Connector/Pin		Voltage	Tolerance
	High	Low		
a.	J1A-36	J1A-37	+ 5.0VDC	± .05 VDC
b.	J1A-38	J1A-39	+10.0VDC	± .05 VDC
c.	J1A-40	J1A-39	-10.0VDC	± .05 VDC
d.	J1A-52	J1A-53	+15.0VDC	± .05 VDC
e.	J1A-54	J1A-53	-15.0VDC	± .05 VDC
f.	J1A-55	J1A-34	+10.0VDC	± .05 VDC

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

a. Preproduction Tests (Performance and Environmental) (See 4.3)
b. Acceptance Tests (See 4.4.1)
c. First Article Inspection (See 4.3.7)

Acceptance of articles under the test provisions specified herein shall not relieve the contractor of responsiblitiy to meet all requirements of the specification.

4.3 Preproduction tests. Preproduction tests shall be conducted by the contractor on equipment representative of the production equipments to be supplied under a production contract. The Government inspector and the procuring activity shall be advised when tests are to be conducted so that a government representative may be designated to witness the tests when so desired. Contractors, not having adequate facilities to conduct all required tests, shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.3.1 Preproduction test procedures. Preproduction test procedures shall be prepared as specified in 4.6.

4.3.2 Preproduction test data. The contractor shall keep a daily record of the equipment, making particular note of any deficiencies or failures.

4.3.3 Scope of tests. Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all requirements of this specification, other applicable specifications and the contract. As a minimum, preproduction tests shall include performance tests and environmental tests.

4.3.3.1 Performance tests. Performance tests shall be conducted at the standard conditions specified in 3.3.9, to verify that operation of the equipment is in accordance with the performance requirements specified herein. The following requirements shall be verified.

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Test	Requirement Paragraph
a. Weight	3.5.1.8
b. Insulation resistance	3.5.3
c. Bonding	3.5.4
d. Continuity	3.5.5
e. Power supply	3.5.6
f. Elevation angle pitch	3.5.7.1
g. Bank angle (roll)	3.5.7.2
h. Heading angle	3.5.7.3
i. Tacan angle	3.5.7.4
j. True airspeed (TAS)	3.5.8
k. UHF homer	3.5.9
l. Air-to-air	3.5.10
m. Air-to-surface	3.5.11
n. Radio altimeter	3.5.12
o. Weapons 'A' mode	3.5.13
p. Invalid ballistic code	3.5.14
q. Pilot's confidence test (PCC)	3.5.15
r. Voltage reference test	3.5.16

4.3.3.2 Environmental tests. Environmental tests shall be conducted under simulated environmental conditions as specified in 3.3.10. The following tests shall be performed.

Test	Requirement Paragraph
a. Vibration	3.3.10.1
b. Temperature/altitude	3.3.10.2
c. Acceleration	3.3.10.3

4.3.3.3 Electromagnetic interference (EMI) test. The contractor shall demonstrate compliance to 3.3.6 by meeting the requirements and performing the tests specified herein. During susceptibility testing, at frequencies where problems are found, the contractor shall determine the threshold level that causes the problem. Tests shall be performed with harnesses that reasonably simulate aircraft configuration as to length, gage, shields, grounding, etc.

4.3.3.3.1 Test requirement and limits of MIL-STD-461. The equipment shall be designed to comply with the following MIL-STD-461 requirements: CE04, CS06, RS02, and RS03 except as noted herein.

4.3.3.3.1.1 Conducted emissions (CE04). Electromagnetic emission, in excess of the broad band limits shown in MIL-STD-461 Figure 5 and the narrow band limits shown in Figure 6 of MIL-STD-461 shall not appear on the signal lines from the WAC between the frequencies of 20 KHz and 50 MHz.

4.3.3.3.1.2 Conducted susceptability (CS06). The equipment shall not exhibit any malfunction or degradation of performance beyond the tolerance given in the equipments individual specification when the voltage spike shown in Figure 10 of MIL-STD-461 is applied to the AC or DC power input lines of the unit under test.

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4.3.3.3.1.3 Radiated susceptability (RS02). The equipment shall not exhibit any malfunction or degradation of performance beyond the tolerance given in the equipments individual specification when subjected to a short duration induction field of amplitude 100V, 10 microsecond spike duration. (E=100 volts across 10 ohms) Figure 10 of MIL-STD-461.

4.3.3.3.1.4 Radiated susceptability (RS03). The equipment shall not exhibit any malfunction or degradation of performance beyond the tolerance given in the equipment individual specification when subjected to a radiated field of 10 volts/meter between the frequencies of 14 KHz to 35 MHz, or 5 volts/meter between the frequencies of 35 MHz to 10 GHz modulated with a 10 microsecond pulse at a 200 PPS rate.

4.3.3.4 Power characteristics test. The equipment shall be tested to demonstrate compliance with the requirement of MIL-STD-704, as specified herein.

4.3.3.4.1 Power consumption test. The power required to operate the equipment shall be measured in the maximum power consumption mode and shall be not greater than the value specified in 3.3.13.

4.3.3.4.2 Power variation test. Power variation tests shall be conducted and equipment shall meet the initiated self-test modes, of 3.5.15 (a and b) during the following simultaneous conditions.

a. Maximum voltage and minimum frequency on all power lines (steady state).

b. Maximum voltage and maximum frequency on all power lines (steady state).

c. Minimum voltage and minimum frequency on all power lines (steady state).

d. Minimum voltage and maximum frequency on all power lines (steady state).

4.3.3.4.3 Under-voltage test. Vary the line voltage continuously from standard conditions to zero and then back to standard conditions over a period of five minutes. Following this test, the equipment shall meet the initiated self-test modes, specified in 3.5.15 (a and b).

4.3.3.5 GSE compatibility demonstration test. Successful completion of the Performance Tests specified in 4.3.3.1 shall demonstrate GSE compatibility.

4.3.3.6 Reliability tests. Reliability Tests shall be performed to evaluate the reliability growth of the equipment. These tests shall consist of 4000 chamber operation-hours of test (3000 hours of equipment 'ON' time) under combined temperature cycling and vibration as outlined in MIL-STD-781, Appendix B. These tests may be performed by similarity when agreed to in advance by the procuring activity.

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4.3.3.6.1 Reliability development. Three equipments shall be subjected to Reliability Development Testing (Test, Analyze and Fix) in accordance with MIL-STD-2068, except as modified below. The purpose of the Reliability Development Test is reliability growth and assessment to promote reliability improvement. Reliability growth is defined as the positive improvement of the equipment reliability through the systematic and permanent removal of failure mechanisms, regardless of their sources, by taking corrective action. The Reliability Development Tests shall be conducted for a minimum duration of 4,000 in-chamber operation (3000 hours equipment 'ON' time) hours. Prior to these tests, the equipment shall have passed the individual performance acceptance tests as specified in 4.3.3.1.

4.3.3.6.2 Test mechanization. The equipment shall be exercised electrically by the test equipment, while being subjected to environmental conditions which simulate aircraft usage. This shall include output of test display signals that represent operational display outputs.

4.3.3.6.3 Temperature. The test chamber shall be programmed to repetitively simulate the AV-8C composite mission profile. The equipment shall be stabilized cold before starting the composite mission cycle. The test chamber shall be capable, as a minimum, of transition rates of 7°C (12°F)/minute for heating and 7°C (12°F)/minute for cooling over the range of -54°C to +71°C.

4.3.3.6.4 Vibration. The vibration on-off periods shall be in accordance with Figure 3. Vibration shall be conducted in accordance with Figure 4.

4.3.3.6.5 Test facilities. Test facilities shall be capable of maintaining specified conditions and of measuring the performance characteristics to the required accuracies. Current calibration of facilities equipment shall be maintained in accordance with MIL-STD-45662.

4.3.3.6.6 Measurements. Equipment performance shall be measured and recorded. Testing shall be discontinued, and required troubleshooting and repair action shall be taken if the value of any specified parameter is not within tolerance. The performance of the equipment shall be monitored during the on-time, either automatically or manually.

4.3.3.6.7 Equipment on-off cycling. Equipment on-off cycling shall be in accordance with Figure 3.

4.3.3.7 Burn-in tests. Burn-in Tests, "as specified in 4.4.1.1.2" shall be performed on each preproduction unit after successfully completing the Performance Tests specified in 4.3.3.1.

4.3.4 Equipment failure. Should a failure occur during preproduction testing, the following action shall be taken:

a. Notify the Government Inspector within one working day.
b. Determine the cause of the failure.
c. Implement and verify corrective action and continue preproduction tests.
d. Repeat any test necessary to verify the corrective action.

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4.3.5 Preproduction test report. The contractor shall prepare for approval by the procuring activity, copies of his preproduction test report. The report shall include the test results, all supporting data collected in conducting the tests and analysis of all failures which occurred. As a supplement the contractor may include, for the procuring activity's consideration, any recommendations for design improvement beyond the corrective action required to pass the test. The data presentation of the test report shall be coordinated with the approved test procedure specified in 4.6.

4.3.6 Preproduction approval. Approval of the preproduction unit shall be by the procuring activity upon satisfactory completion of all tests.

4.3.7 First article inspection. The preproduction tests listed herein meet the contractual requirements that may be specified for a first article inspection.

4.4 Production equipments. Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equal to the approved preproduction units. Each equipment shall be capable of sucessfully passing the same tests as imposed on preproduction units. Evidence of noncompliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.4.1 Acceptance tests. The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance test. All inspection and testing shall be under the observation of the Government Inspector. Contractors not having adequate facilities for conducting all required tests shall engage the services of a commercial testing laboratory acceptable to the procuring activity. The contractor shall prepare test reports showing results for acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval, of material during the course of manufacture, shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

a. Individual Tests
b. Special Tests

4.4.1.1 Individual tests. Each equipment submitted for acceptance shall be subjected to the Individual Tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

a. Examination of Product
b. Burn-In Test
c. Performance Tests

4.4.1.1.1 Examination of product. Each equipment shall be examined carefully to determine the material and workmanship requirements have been met, and physical characteristics of the equipment agree with the outline dimensions specified herein.

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4.4.1.1.2 Burn-in-test procedure. Each equipment shall be operated under the conditions specified for a period of 15 in chamber hours without a relevant failure. Each equipment shall be subjected to six test cycles (each cycle 2.5 hours) of temperature cycling from -54°C to +60°C including power ON/OFF periods and random vibration (without vibration isolators) at 3.25g's for 10 minutes at the temperature extremes. Random vibration test periods are cumulative and may be performed during any portion of the manufacturing burn-in test as long as the complete vibration requirement of 60 minutes at each temperature extreme is fulfilled. At any one time this shall not exceed 1 months production and shall be with the concurrence of the procuring activity. The test cycle may be extended beyond the specified duration to complete the random vibration requirement. Reference Figure 3 and Figure 4.

A relevant failure shall be defined as a degration of a function beyond the specified limits as described in the performance test specified in 4.3.3.1. If a relevant failure occurs during the burn-in-test, corrective action shall be implemented to repair the fault and the entire 15 hour period of manufacturing burn-in repeated.

4.4.1.1.3 Performance test. Each equipment shall be operated long enough to permit the equipment temperature to stabilize. The requirements specified in 4.3.3.1 shall be verified to assure satisfactory equipment operation.

4.4.1.2 Special tests. Special tests shall be conducted when required by the procuring activity, (see 6.2) on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control.

4.4.1.2.1 Special test schedule. Selection of equipments for special tests shall be made as follows:

a. On an early equipment after an engineering or material change.

b. Whenever failure reports or other information indicate that additional tests are required. (This will be determined by the procuring activity.)

4.4.1.2.2 Scope of tests. Special tests shall consist of such tests as approved by the procuring activity. Test procedures, previously approved for the first article tests, shall be used where applicable. When not applicable, the contractor shall prepare a test procedure for procuring activity approval prior to conducting the tests.

4.4.1.2.3 Equipment failure. Should a failure occur during the special tests, the following action shall be taken:

a. Notify the Government Inspector within one working day.

b. Determine the effectivity of BIT to detect and isolate the failure.

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c. Determine the cause of the failure.

d. Determine if the failure is due to a material or a design defect.

e. Prepare for the procuring activity, proposed corrective action.

f. Where practical, include a test in the Individual Test to check all equpment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.4.1.3 Reliability assurance warranty. When authorized by the procuring activity, a reliability warranty shall apply to the equipment specified in the warranty agreement. The reliability warranty provisions shall be detailed in a document attached to and made a part of the contract.

4.4.1.4 Repair service contract. When authorized by the procuring activity, maintenance and repair services to test the equipment, analyze faults, repair or replace defective parts, reassemble, retest to performance standards, and ship as directed, shall be performed by the contractor. The provisions of this service, and the equipment affected, shall be detailed in a document attached to and made a part of the contract. (See 6.2).

4.4.2 Equipment failure. Should a failure occur during production testing, the following action shall be taken:

a. Notify the Government Inspector within one working day.

b. Determine the cause of the failure.

c. Prepare corrective action for Government Inspector information.

d. Implement and verify corrective action and continue production tests.

e. Repeat any tests necessary to verify the corrective action.

4.5 Test data. The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failure, the defective part shall be replaced and operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test. This record shall indicate the following:

a. Part type number

b. The circuit reference symbol number

c. The part function

d. Name of the manufacturer

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e. Nature of the failure

f. The number of hours which the part operated prior to failure

4.6 First article test procedures. The procedures, used for conducting first - article tests and acceptance tests, shall be prepared by the contractor for procuring activity review and approval. MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when their use is approved by the procuring activity.

4.7 Reconditioning of tested equipment. Equipment, which has been subjected to environmental tests as described in 4.3.3.2, shall be assessed for refurbishment to a flight-worthy condition. Upon approval and funding by the procuring activity, the equipment shall be reconditioned by the contractor and shall be resubmitted for acceptance.

4.8 Presubmission testing. No WRA shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, with all applicable requirements.

4.9 Rejection and retest. Equipment, which has been rejected, may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and action taken to correct the defects found in the original shall be made available to the Government Inspector.

5. PACKAGING

5.1 Packaging Requirements. All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract.

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6. NOTES

6.1 Intended use. The equipment covered by this specification should perform as an integral part of the Weapon Aiming System which provides weapon aiming information for Air-to-Air gun lead computation, Air-to-Surface bombs impact computation, Air-to-Surface rockets and guns impact computation, generation of a flashing breakaway warning symbol for ground avoidance, and interface of UHF Homer Signal for use by Attitude Director Indicator (ADI).

6.2 Ordering data.

6.2.1 Acquisition requirement. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

a. Title, number, and date of this specification

b. Selection of applicable levels of packaging and packing (see 5.1).

c. Special tests if required (See 4.4.1.2)

6.2.2. Data requirements. When this specification is used in an acquisition which incorporate a DD Form 1423, Contract Data Requirements List (CDRL) the data requirements identified below shall be developed as specificed by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104-9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph no.		Data requirements	Applicable DID no.
(a)	4.3.5	Preproduction Test Reports	UDI-E-23923
(b)	4.4.1	Acceptance Test Reports	UDI-E-26401
(c)	4.5	Failure Reports	DI-R-7041

6.3 First article. When a first article inspection is required, the item will be tested and should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test approval of the documents first article.

6.4 Associated equipment. The equipment is compatible with the other components of the Digital Data Display Set AN/AVQ-30(V). The equipment should meet all requirements of this specification when installed in the AV8C aircraft and operating in conjunction with the Digital Data Display Set and the associated equipment. The weapon aiming system should comprise the following equipment:

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NOMENCLATURE

a. Weapon Aiming Computer CP-1444/A
b. Digital Data Display Set AN/AVQ-30 (V)*
c. Single Axis Rate Ryro, 405 RGU
d. Twin Axis Rate Gyro, 307 RGU
e. Air Data Computer, 1202 KHA/1
f. Vertical Speed Transducer
g. Airstream Direction Detector, FE51-56280, B282634
h. UHF Homer Adaptor, 705483-801 MAGNAVOX
i. Camera Control Panel (Target Height Set Potentiometer), B310828
j. Ballistics Selector Panel
k. Weapon Control Panel, 700-1-17907, 701-1-17911
l. TACAN Converter, 8010000093-3
m Electronic Control Amplifier (SOARS)
n. Weight on Wheels Microswitch
o. Control Column (Bomb and Gun Safety Switches)
p. Centralized Warning Panel, P-B282717 S-B282726
q. Throttle Twist Grip, C305689
r. Sidewinder Control Panel
s. Center Console WAC Test Switch

*Includes the following equipment:
Signal Data Converter, CV-3600/AVQ-30 (V)
Digital Display Indicator, IP-1351/AVQ-30 (V)
Display Set Control, C-10626/AVQ-30 (V)
Power Supply PP-7553/AVQ-30 (V)
Video, Amplifier AM-7032/AVQ-30 (V) **
Indicator Control C-10627/AVQ-2-30 (v) **

**TAV8 - Only

Preparing Activity
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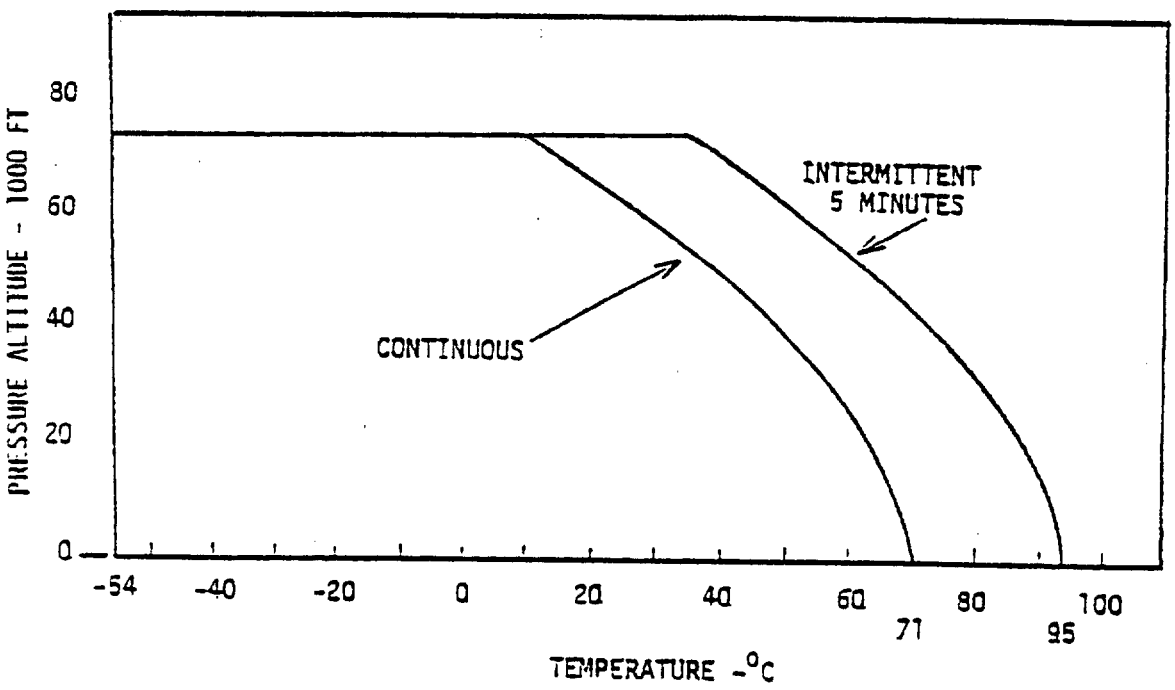
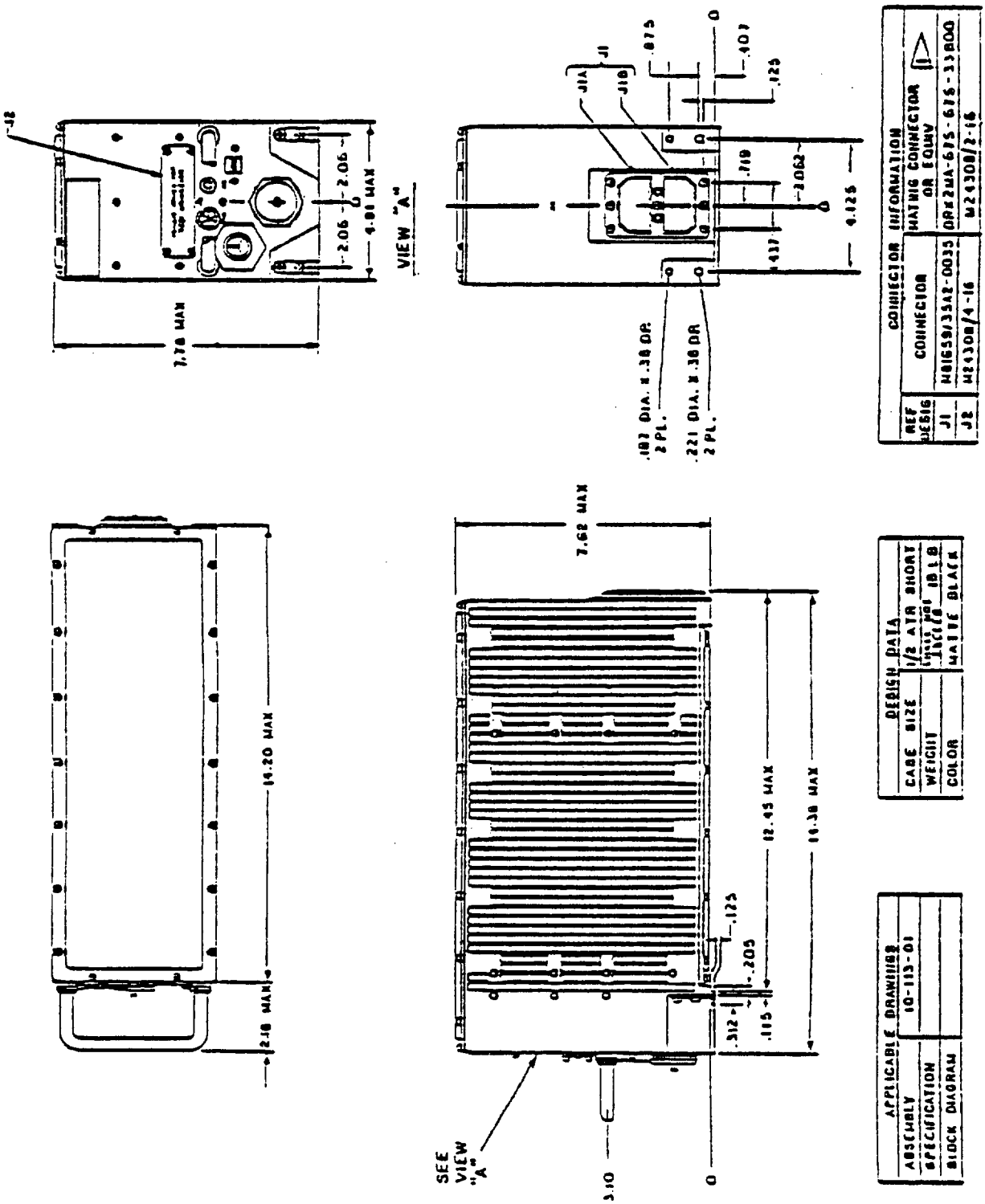


FIGURE 1. Temperature altitude conditions (class 2 equipment).

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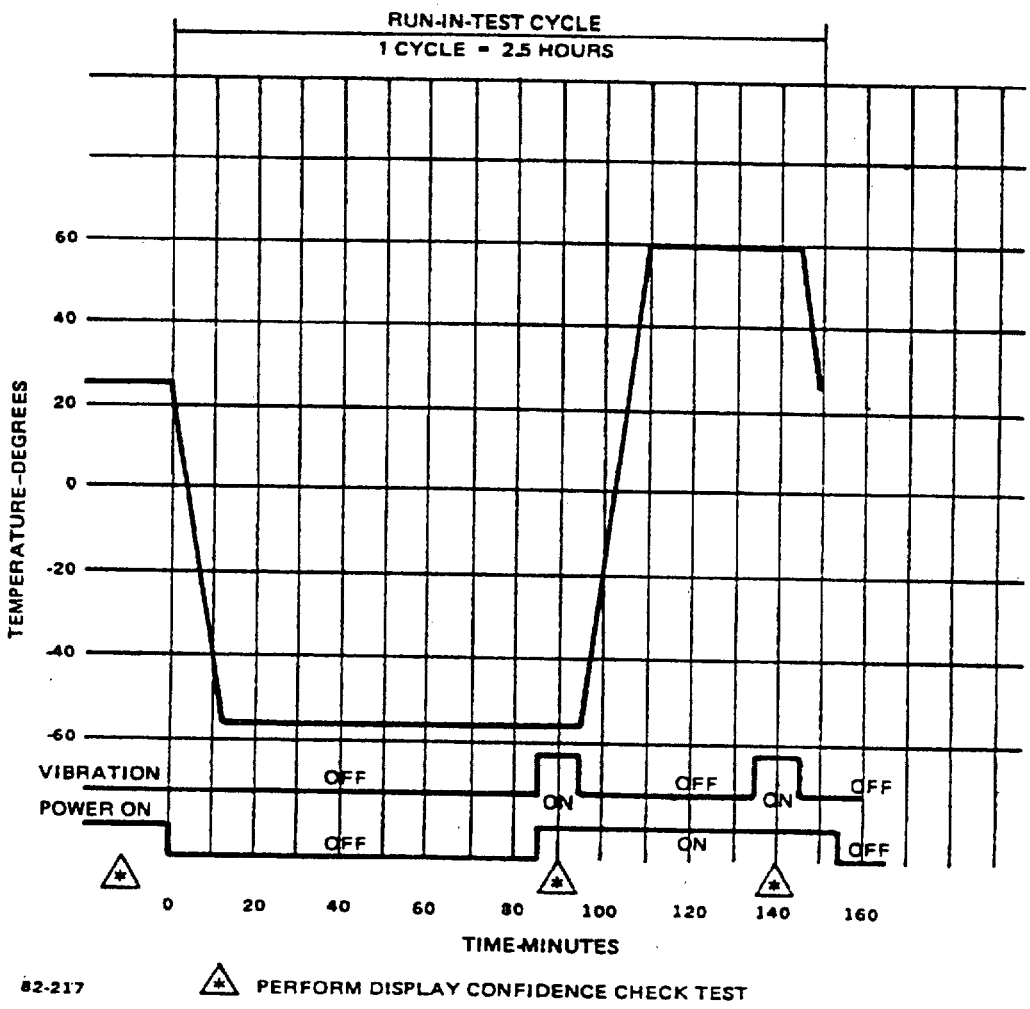


FIGURE 3. Burn-in test cycle.

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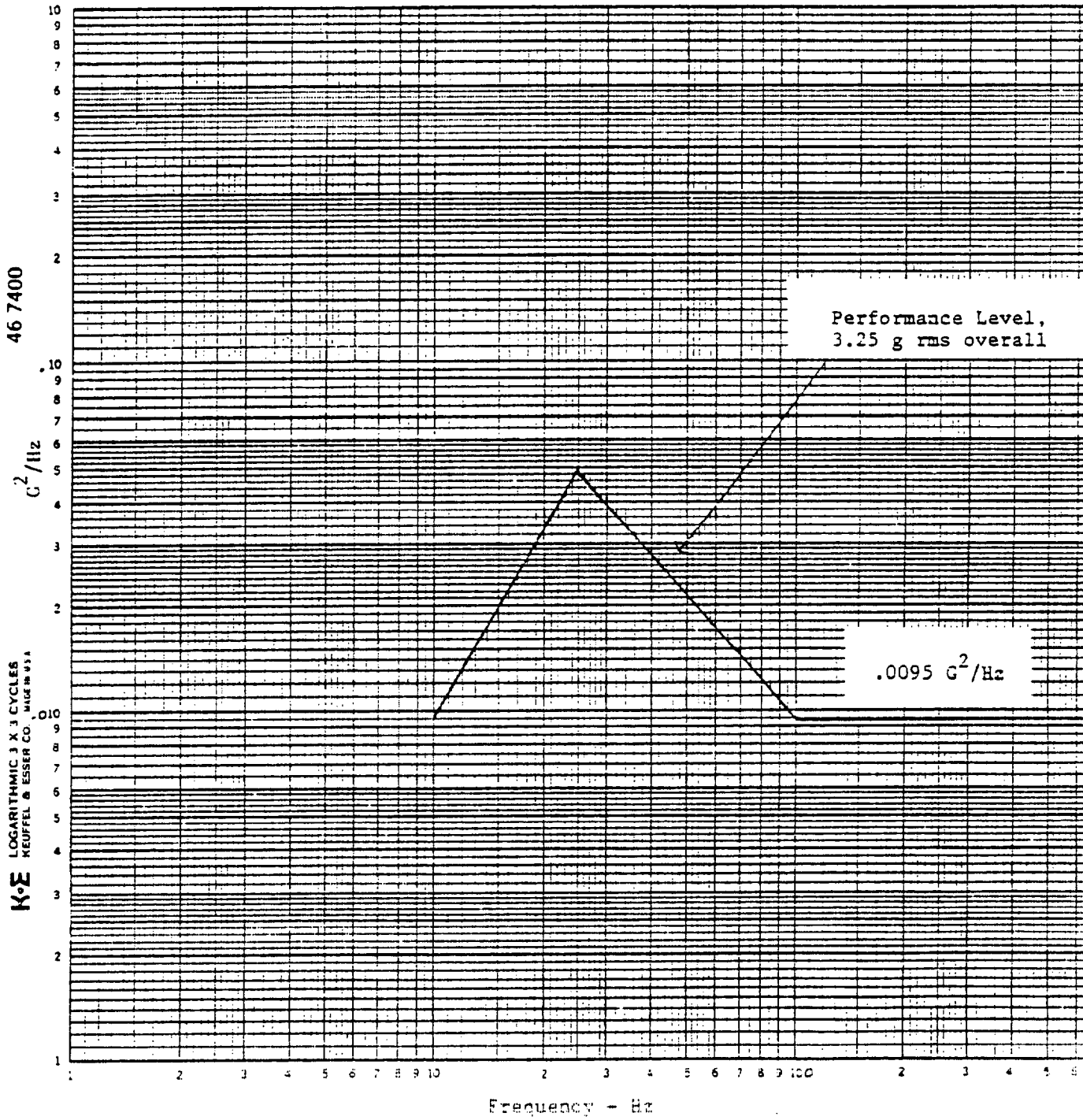


FIGURE 4. Vibration test profile.

NOTE: Reference Paragraph 4.4.1.1.2 that a tolerance of +3.5dB, -9dB is permitted above 200 Hz to permit multiple unit burn-in to be accomplished.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

MIL-C-85338(AS) Computer Weapon Aiming CP-1444/A

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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